

Cross-lingual Transfer between Azerbaijani and Turkish: Separating Syntactic Transfer, Lexical Warm-Up, and Tokenizer Quality

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Abstract—Two routes are commonly proposed for adapting a multilingual encoder to a lower-resource language: intermediate training on a closely related high-resource language, and replacing the tokenizer with a language-specific one whose embeddings are reconstructed without training. We compare both for binary Azerbaijani sentiment classification on XLM-15 (Azerbaijani absent from pretraining) and XLM-R (Azerbaijani present) under a hash-locked, pre-registered protocol with a fixed 2,000-step fine-tuning budget. Seven conditions—direct fine-tuning, orthogonal-matching-pursuit (OMP) transplantation of an Azerbaijani donor tokenizer, two placebo initialisers, real Turkish intermediate training, its word-shuffled control, and their combination—were run at $n \in \{500, 2,000\}$ with five seeds, plus a higher-data stress test, for 164 completed runs. Because classifier collapse is frequent for XLM-15, we report escape probability, macro-F1 conditional on escape, and all-seed macro-F1 separately; every test score comes from the restored, verified validation-selected checkpoint. The clearest result is negative: for XLM-R the implemented OMP transplant lowers macro-F1 from 0.8277 to 0.7618 on validation at $n = 2,000$ and from 0.7970 to 0.7404 on test, in every seed and at both sizes, and intrinsic diagnostics (masked-token top-1 accuracy 36.9%→2.7%, +5.34 bits per character) corroborate the failure. Turkish intermediate training is numerically best for XLM-R but its gains over direct fine-tuning (+0.0169 at $n = 500$, +0.0030 at $n = 2,000$) do not survive Holm correction over the 86-test family; the 27 contrasts that do survive, all on XLM-R, keep their direction and significance on test, subject to the five-seed exact sign-flip floor of $p = 0.0625$. XLM-15 results are dominated by collapse (60/94 escapes) and by 14 of 30 degenerate Turkish source stages. Code, configuration, and all 164 run records: <https://github.com/shahinsafarli/az-tokenizer-project> (tag v1.0-final).

Index Terms—Azerbaijani NLP, cross-lingual transfer, low-resource languages, tokenizer transplantation, orthogonal matching pursuit, intermediate-task training, multilingual Transformers, experimental methodology, pre-registration

I. INTRODUCTION

Multilingual encoders such as XLM [1] and XLM-R [2] are the default starting point for text classification in languages with little labelled data. Azerbaijani is a representative case: a Turkic language written in Latin script with roughly ten million speakers, a small labelled-data ecosystem, uneven coverage in multilingual pretraining corpora [3], and a close, high-resource relative—Turkish—with which it shares much vocabulary and morphology. Two adaptation strategies therefore suggest themselves. The first is *intermediate training* on Turkish before Azerbaijani fine-tuning, in the spirit of STILTs [4], [5]; Kardeş-NLU [6] reports that such a “high-resource cousin” stage helps Turkic targets. The second is *tokenizer transplantation*: replacing the encoder’s vocabulary with one trained for the target language and initialising the new embedding rows without gradient updates, most recently by orthogonal matching pursuit (OMP) over shared anchor tokens [7], following a line of embedding-initialisation work [8]–[11].

A. The identification problem

A gain after Turkish training would not by itself reveal *why* Azerbaijani performance changed. It could reflect order-sensitive syntactic transfer, shared sentiment-bearing vocabulary, extra task optimisation on a larger corpus, or a different interaction with the pretrained tokenizer. Tokenizer replacement is a second compound intervention: it changes segmentation efficiency, the vocabulary, the embedding table, the truncation pattern, the tied output rows, and the total parameter count at once. Two base encoders add a third layer of confounding, because they differ in architecture, capacity, and pretraining-language coverage. A study that reports one accuracy number per pipeline can rank the pipelines but cannot attribute the

ranking to a mechanism. This paper is therefore deliberately titled a *comparison*: it measures what each implemented pipeline does under a fixed protocol, states which contrasts the design identifies, and refuses the mechanistic claims it cannot support.

B. Research questions and hypotheses

Under a fixed 2,000-optimizer-step Azerbaijani fine-tuning protocol, we ask how Turkish intermediate training and Azerbaijani donor-tokenizer initialisation compare in

- (i) the probability of escaping classifier collapse (a constant-prediction basin at macro-F1 = 1/3),
- (ii) validation macro-F1 conditional on escape and over all seeds, and
- (iii) test macro-F1 of the restored validation-selected checkpoint.

Two conditional hypotheses were fixed before the grid ran:

- **H1 (order-sensitive transfer).** If Turkish intermediate training helps through order-sensitive structure rather than lexical overlap or extra optimisation alone, real Turkish training will outperform a word-shuffled Turkish control that preserves every example’s word multiset and label.
- **H2 (segmentation bottleneck).** If XLM-15’s poor Azerbaijani segmentation is a binding bottleneck *and* OMP-reconstructed embeddings are usable, donor-tokenizer initialisation will help XLM-15 more than XLM-R, whose pretraining already covers Azerbaijani.

Neither hypothesis licenses a single-mechanism causal claim; Section III makes the identification limits explicit.

C. Contributions

- 1) A completed, pre-registered 164-run comparison of seven adaptation pipelines across two encoders, two primary data sizes, and a higher-data stress test, with zero failed runs and every number traceable to a committed artefact (Section V).
- 2) Failure-aware reporting: escape probability with Wilson intervals, conditional and all-seed macro-F1 with bootstrap intervals, never pooled, plus training trajectories that show *when* collapse happens (Section VI-D).
- 3) A multiplicity-controlled inference family (86 Welch contrasts, Holm-corrected), a test-side confirmation restricted to the 27 validation survivors, and an exact sign-flip analysis that states the distribution-free floor of a five-seed design (Sections VI-E–VI-G).
- 4) Intrinsic transplant diagnostics—anchor identification modes, masked-token top-1 accuracy, bits per character, embedding-norm calibration, and bit-exact fidelity controls—that independently corroborate the downstream failure of the implemented OMP transplant on XLM-R (Section VI-B).
- 5) Source-stage degeneracy accounting for Turkish intermediate training (14 of 30 XLM-15 stages degenerate), an explicit audit of data provenance, label quality, software, compute, and test-set usage, and a claim-by-claim evidence map (Appendix E).

D. Organisation

Section II reviews related work. Section III formalises the conditions, estimands, and identification map. Section IV describes anchor identification, OMP transplantation, the placebo initialisers, the Turkish stage and its control, the training procedure, and the statistical plan. Section V documents data, pre-registration, the grid, the test protocol, compute, and reproducibility. Section VI presents tokenizer-level, intrinsic, and downstream results. Sections VII and VIII interpret the evidence and its limits; Section IX concludes. Appendices give the complete 36-cell results, all contrasts, the frozen configuration, and the evidence map.

II. BACKGROUND AND RELATED WORK

A. Multilingual encoders and cross-lingual transfer

XLM introduced masked and translation language modelling over multiple languages [1]; its 15-language MLM+TLM checkpoint fine-tuned on XNLI [12] is the primary encoder here because its documented pretraining list includes Turkish but not Azerbaijani. XLM-R scaled the recipe to 100 languages, including Azerbaijani, with a 250k SentencePiece vocabulary [2], [13]. Multilingual BERT-style models transfer zero-shot across languages to a degree that depends on typological similarity and script [14], [15], but low-resource languages are systematically under-served both in pretraining data and in vocabulary allocation [3]. Our two bases were chosen to bracket this axis: one without and one with Azerbaijani pretraining exposure.

B. Transfer through a related high-resource language

Supplementary training on intermediate labelled tasks (STILTs) can improve downstream fine-tuning, though the benefit is task- and data-dependent [4], [5]. Kardeş-NLU studies exactly the Turkic setting: transferring from Turkish to lower-resource cousins including Azerbaijani, and finds that intermediate Turkish fine-tuning helps especially when target supervision is scarce [6]. That result motivated our Turkish stage and our $n = 500$ cells. Kardeş-NLU does not, however, separate the contribution of word order from that of shared vocabulary and additional optimisation; our word-shuffled control (Section IV-G) is a first step toward that separation, and we are explicit about how far it goes.

C. Tokenization and under-represented languages

Subword tokenizers [13], [16] allocate vocabulary in proportion to pretraining corpus frequency, so languages with little data are segmented into many short pieces. Rust et al. show that this “fertility” gap materially affects monolingual downstream performance of multilingual models [17]; Ahia et al. quantify the cost asymmetry across languages [18]; Toraman et al. analyse tokenization choices for Turkish specifically [19]. Our Section VI-A measures the gap directly: XLM-15 needs 3.21 tokens per Azerbaijani word against 1.90 for XLM-R and 1.76 for the Azerbaijani donor, and its vocabulary contains no token bearing the letters ş, ç, ö, ü or ğ.

D. Tokenizer transfer and embedding initialisation

Replacing a model’s tokenizer requires initialising embeddings for tokens the model has never seen. Vocabulary augmentation with new rows [20], WECHSEL’s alignment through bilingual static embeddings [8], FOCUS’s sparse combinations of overlapping tokens [9], and fast vocabulary transfer for compression [10] all initialise new rows from old ones. Zero-Shot Tokenizer Transfer learns a hypernetwork that predicts embeddings for arbitrary tokenizers [11]. The training-free OMP method we evaluate [7] expresses each unseen donor token as a k -sparse combination of anchor tokens shared with the base vocabulary, solved in the donor’s embedding space, and transfers the *coefficients* to the base space; OMP itself is a classical greedy sparse-approximation algorithm [21], [22]. Our implementation follows this recipe with a candidate restriction and a norm-rescaling step, and we treat mean and random-coefficient initialisers as placebos. We evaluate one Azerbaijani application and do not claim to reproduce every assumption of the precursor method.

E. Azerbaijani NLP resources

Open foundation models for Azerbaijani [23] and the HPLT multilingual corpora and encoders [24] show that language-specific resources are becoming available; the HPLT Azerbaijani BERT [25] supplies our donor tokenizer and donor embedding space. The Azerbaijani sentiment data come from a public Hugging Face dataset [26], the Turkish source data from TRSAv1 [27].

F. Statistical practice in fine-tuning experiments

Fine-tuning pretrained encoders is unstable across random seeds and can collapse to degenerate solutions [28], [29]; benchmark variance is dominated by such sources unless multiple seeds are run [30]. Dror et al. recommend explicit significance testing and multiplicity awareness in NLP [31]. We follow that advice with Holm’s step-down procedure [32], Wilson intervals for escape rates [33], Welch’s t for unequal variances [34], percentile bootstrap intervals [35], and exact permutation reasoning [36] to state what five seeds can and cannot establish.

III. PROBLEM FORMULATION

A. Notation

Let $\mathcal{B} = \{\text{XLM-15}, \text{XLM-R}\}$ be the base encoders, each with an original tokenizer τ_B and input-embedding matrix $\mathbf{E}_B \in \mathbb{R}^{V_B \times d_B}$ ($V_B = 95,000$, $d_B = 1024$ for XLM-15; $V_B = 250,002$, $d_B = 768$ for XLM-R). The donor is the HPLT Azerbaijani BERT with tokenizer τ_D and embeddings $\mathbf{E}_D \in \mathbb{R}^{V_D \times d_D}$, $V_D = 32,770$. A *condition* $c = (\kappa, \pi)$ combines a tokenizer/initialisation choice $\kappa \in \{\text{original}, \text{donor/OMP}, \text{donor/mean}, \text{donor/random}\}$ with a pre-Azerbaijani stage $\pi \in \{\text{none}, \text{real TR}, \text{shuffled TR}\}$. Seven of the twelve combinations were run (Table I); the remaining five (placebo initialisers with a Turkish stage, and donor tokenizer with shuffled Turkish) were frozen out of scope before the grid to keep the run count inside the compute budget, and their absence

TABLE I
THE SEVEN CONDITIONS. IDS AND NAMES ARE THOSE OF CONFIGS/EXPERIMENT.YAML; “SIZES” LISTS THE AZERBAIJANI TRAINING SIZES AT WHICH THE CONDITION WAS RUN FOR EACH BASE.

Short	Config name (id)	Init. ^b	Pre-AZ	Sizes
DIRECT	baza (1)	–	none	all ^a
OMP	tokenizator (3)	OMP	none	all ^a
MEAN	transplant_mean (6)	mean	none	all ^a
RANDOM	transplant_random_coef (7)	random	none	all ^a
TR	turk (2)	–	real TR	500, 2000
OMP+TR	her_ikisi (4)	OMP	real TR	500, 2000
SHUF.TR	turk_qarisiq (5)	–	shuf. TR	500, 2000

^aXLM-15: 500, 2000, 10000, 20914; XLM-R: 500, 2000. ^b“–” keeps the original tokenizer; the other three install the AZ donor tokenizer initialised by OMP ($k=64$), the mean of the k nearest anchors, or random coefficients (seed 0).

is listed as a limitation. A run is (B, c, n, s) with Azerbaijani training size n and seed s .

B. Estimands

Let $F(t)$ denote validation macro-F1 of a run at optimizer step $t \in \mathcal{T} = \{200, 400, \dots, 2000\}$. A constant predictor on the balanced binary validation split scores $F = 1/3$; we call this the *collapse basin*. A run *escapes* if

$$E = \mathbf{1} \left[\max_{t \in \mathcal{T}} F(t) > \theta \right], \quad \theta = 0.40, \quad (1)$$

where θ was fixed from pilot runs and hash-locked before the grid. The selected checkpoint is the earliest maximiser

$$t^* = \min \{t \in \mathcal{T} : F(t) = \max_{t'} F(t')\}, \quad (2)$$

and every run reports $F_{\text{val}} = F(t^*)$ and, after restoring and numerically verifying the checkpoint, F_{test} on the fixed test split. For a cell (base, condition, n) with seeds $\mathcal{S}$ we report three quantities that answer different questions and are never pooled:

$$\hat{p}_{\text{esc}} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} E_s, \quad (3)$$

$$\bar{F}_{\text{cond}} = \frac{1}{\sum_s E_s} \sum_{s \in \mathcal{S}} E_s F_s, \quad (4)$$

$$\bar{F}_{\text{all}} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} F_s. \quad (5)$$

$\bar{F}_{\text{cond}}$ describes predictive quality *given* that optimisation succeeded; it is a selection on outcome and can compare disjoint survivor sets across conditions. $\bar{F}_{\text{all}}$ charges every collapsed seed at its actual score and is the quantity a practitioner would experience without re-running seeds. macro-F1 is the unweighted mean of the per-class F1 scores,

$$\text{macro-F1} = \frac{1}{2} \sum_{y \in \{0,1\}} \frac{2 \text{TP}_y}{2 \text{TP}_y + \text{FP}_y + \text{FN}_y}. \quad (6)$$

C. Hypotheses as contrasts, and what they identify

Each hypothesis maps to within-base contrasts on the estimands above. Table II states, for each contrast, what changes jointly and therefore what the contrast can and cannot isolate. Cross-base comparisons are descriptive only: the two encoders differ in architecture, non-embedding capacity (151.7M vs. 86.0M parameters), and pretraining coverage, so a difference

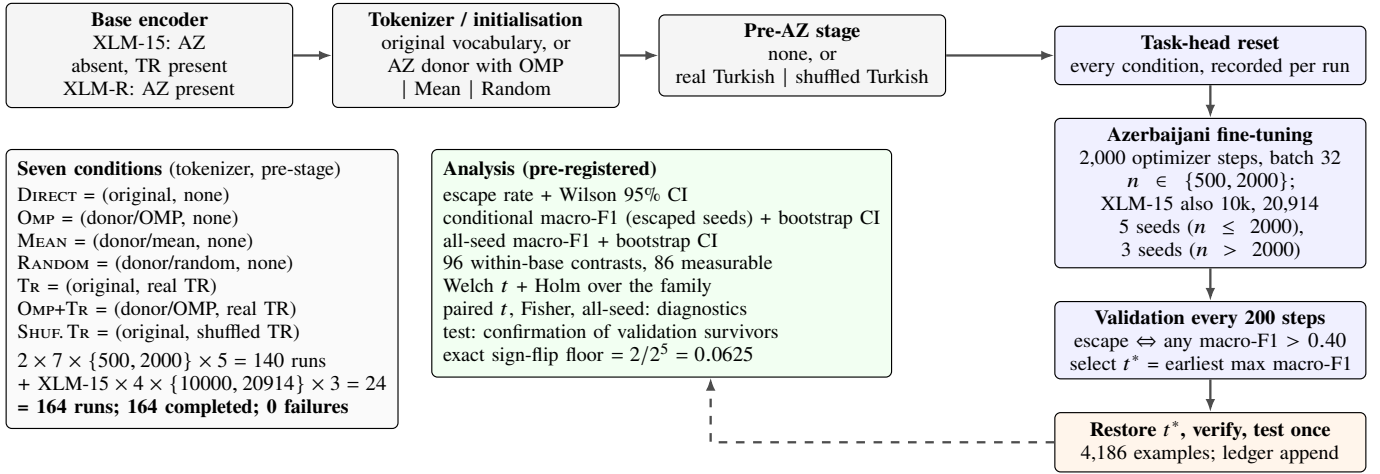


Fig. 1. Experimental design. Each run is a triple (base, condition, seed) at one Azerbaijani training size n ; a condition is a pair (tokenizer/initialisation, pre-Azerbaijani stage). Every arm passes through the same task-head reset, the same 2,000-step Azerbaijani budget, the same validation cadence and selection rule, and exactly one test evaluation of the restored selected checkpoint. Counts are from the frozen configuration (evidence/preregistration/FROZEN.md) and the launcher record (results/launcher_state.json).

TABLE II

IDENTIFICATION MAP: WHAT EACH PRE-REGISTERED CONTRAST CHANGES JOINTLY AND WHAT IT CAN THEREFORE SUPPORT. “ISOLATES” IS INTENTIONALLY STRICT.

Contrast	Jointly changed	Supports / does <i>not</i> isolate
DIRECT → OMP	vocabulary, segmentation, embedding rows, tied output rows, truncation, parameter count	net effect of the implemented transplant pipeline; <i>not</i> tokenizer quality vs. embedding quality
OMP → MEAN, OMP → RANDOM	only the coefficient solver on the same donor vocabulary and anchors	whether the fitted sparse solve beats a crude or random weighting; <i>not</i> whether any initialiser is adequate
DIRECT → TR	source-task exposure, extra optimisation, lexical overlap, word order	net effect of Turkish intermediate training; <i>not</i> its mechanism
SHUF. TR → TR	word order, fluency, source-task difficulty (multiset and labels preserved)	order-sensitive component (H1); <i>not</i> syntax alone
DIRECT → SHUF. TR, OMP → OMP+Tr, TR → OMP+Tr	source exposure without natural order interaction of transplant and Turkish stage	lexical/optimisation component under a degraded source task whether the transplant offsets the Turkish outcome; <i>not</i> whether Turkish knowledge transferred through donor rows
XLM-15 vs. XLM-R	architecture, capacity, pretraining coverage, tokenizer scheme	descriptive contrast for H2; <i>not</i> the effect of Azerbaijani pretraining exposure

between bases cannot be attributed to Azerbaijani pretraining exposure alone.

IV. METHOD

A. Base encoders and donor

Table III lists the three models. The primary encoder is the 15-language XLM MLM+TLM checkpoint fine-tuned on XNLI (FacebookAI/xlm-mlm-tlm-xnli15-1024), whose

documented pretraining languages include Turkish but not Azerbaijani; it uses a 95k fastBPE vocabulary with lower-casing and accent removal, which is why its vocabulary contains no token with the letters ş, ç, ö, ü or ğ (Section VI-A). The contrast encoder is `xlm-roberta-base`, pretrained on 100 languages including Azerbaijani with a 250k SentencePiece vocabulary. The donor is `HPLT/hplt_bert_base_az`, pinned at revision `a126552c3073`, a byte-level SentencePiece model with 32,770 tokens trained on Azerbaijani.

Total parameter counts conflate the vocabulary-dependent embedding table with the vocabulary-independent encoder, so we decompose them. Transplantation leaves every encoder layer, attention and feed-forward weight, layer norm, and (for XLM-15) position and language embedding bit-for-bit unchanged; the 25.6% and 60.0% total-parameter reductions are entirely the smaller embedding table. The two *encoders* are not capacity-matched (151.7M vs. 86.0M non-embedding parameters, a 1.8× factor), which is one reason cross-base contrasts are descriptive.

B. Anchor identification

Transplantation needs a set of *anchors*: donor tokens that also exist in the base vocabulary, so that their donor- and base-space embeddings are both available. Three identification modes were measured (Table IV). *Strict* intersects canonical token strings including word-boundary markers; *surface* intersects boundary-stripped strings and can mis-pair a word-final donor piece with a word-initial base piece; *functional*, the pre-registered default, asks the base tokenizer itself. For each donor token v with stripped surface $\sigma(v)$,

$$\mathcal{A} = \left\{ (v, u) : \text{enc}_{\tau_B}(\sigma(v)) = \langle u \rangle, u \notin \mathcal{U}_B \right\}, \quad (7)$$

where enc_{τ_B} encodes without special tokens and $\mathcal{U}_B$ is the set of unknown and special ids. The exclusion of $\mathcal{U}_B$ matters: a surface the base tokenizer cannot represent encodes to exactly one token, `<unk>`, and without the exclusion such

TABLE III

MODELS. PARAMETER COUNTS MEASURED FROM THE LOADED ARCHITECTURES (HANDOFF, MD §14); EMBEDDING PARAMETERS ARE THE INPUT TOKEN TABLE ONLY.

	XLM-15	XLM-R	AZ donor
Hugging Face id	xlm-mlm-tlm-xnli15-1024	xlm-roberta-base	HPLT/hplt_bert_base_az
Role	primary	contrast	donor
Azerbaijani in pretraining	no	yes	yes (only)
Turkish in pretraining	yes	yes	no
Tokenizer scheme	fastBPE	SentencePiece	SentencePiece, byte
Vocabulary	95,000	250,002	32,770
Hidden size	1024	768	768
Embedding params (original)	97,280,000	192,001,536	—
Embedding params (transplanted)	33,556,480	25,167,360	—
Non-embedding params	151,698,434	86,043,650	—
Total (original / transplanted)	249.0M / 185.3M	278.0M / 111.2M	—
Licence	CC-BY-NC-4.0	MIT	Apache-2.0

Algorithm 1: Functional anchor identification (src/tokenization/anchor_map.py)

Input: base tokenizer τ_B with invalid ids $\mathcal{U}_B = \{\text{unk}\} \cup \text{special}$; donor vocabulary $\{(v, \text{str}_v)\}$
Output: anchor pairs $\mathcal{A}$; unfamiliar donor ids $\mathcal{V}_{\text{unf}}$

```

1  $\mathcal{A} \leftarrow \emptyset$ ;
2 foreach donor id  $v$  in ascending order do
3    $\sigma \leftarrow$  decoded surface of  $\text{str}_v$ , boundary marker stripped;
4   if  $\sigma = \emptyset$  then continue;
5   foreach  $x \in (\sigma, \text{"_"} + \sigma)$  do
6      $\mathbf{u} \leftarrow \text{enc}_{\tau_B}(x)$  without special tokens;
7     if  $|\mathbf{u}| = 1$  and  $u_1 \notin \mathcal{U}_B$  then
8        $\mathcal{A} \leftarrow \mathcal{A} \cup \{(v, u_1)\}$ ; break;
9  $\mathcal{V}_{\text{unf}} \leftarrow \{v : (v, \cdot) \notin \mathcal{A}\} \setminus \text{special}$ ;
10 return  $\mathcal{A}, \mathcal{V}_{\text{unf}}$ 

```

pairs would bind donor rows to the UNK vector and pull every reconstruction toward it (Amendment 8.2 in the frozen specification). The standalone surface is tried first, then the surface with a leading space; both variants are counted, and in the final evidence all matches were standalone. Algorithm 1 states the procedure. Seven special-token rows are aligned separately and added to the anchor set at installation time, giving 8,054 and 9,783 mapped rows for 8,047 and 9,776 lexical anchors.

C. OMP embedding reconstruction and coefficient transfer

Let $\mathbf{D}_{\mathcal{A}} \in \mathbb{R}^{|\mathcal{A}| \times d_D}$ and $\mathbf{B}_{\mathcal{A}} \in \mathbb{R}^{|\mathcal{A}| \times d_B}$ hold the donor- and base-space embeddings of the anchors, with $\tilde{\mathbf{d}}_a = \mathbf{d}_a / \|\mathbf{d}_a\|_2$ the ℓ_2 -normalised donor rows. For an unfamiliar donor token v with normalised embedding $\tilde{\mathbf{d}}_v$, the candidate set is the $m = 256$ anchors with the highest donor-space cosine similarity, sorted in decreasing order,

$$C(v) = \text{top}_m \{a \in \mathcal{A} : \tilde{\mathbf{d}}_v^\top \tilde{\mathbf{d}}_a\}, \quad (8)$$

and the coefficients solve the k -sparse least-squares problem

$$\alpha_v = \arg \min_{\alpha \in \mathbb{R}^m} \|\tilde{\mathbf{d}}_v - \tilde{\mathbf{D}}_{C(v)}^\top \alpha\|_2^2 \quad \text{s.t.} \quad \|\alpha\|_0 \leq k, \quad k = 64, \quad (9)$$

approximately, by orthogonal matching pursuit [21], [22] (scikit-learn’s `OrthogonalMatchingPursuit`, no inter-

cept). The coefficients, not the vector, are transferred: the base-space row is

$$\hat{\mathbf{b}}_v = \sum_{a \in C(v)} \alpha_{v,a} \mathbf{b}_a, \quad (10)$$

using the *unnormalised* base anchor rows, so that $\hat{\mathbf{b}}_v$ lives on the base model’s own scale and $d_D \neq d_B$ is permitted. Restricting to $m = 256 < d_D = 768$ keeps the dictionary undercomplete, which the implementation warns is required for greedy selection to be reliable. The candidate sorting in (8) is what makes the mean placebo below well defined (Amendment 8.1).

D. Placebo initialisers and norm rescaling

Two placebos use the same donor vocabulary, anchors, and candidate sets but replace the solver:

$$\alpha_{v,a}^{\text{mean}} = \frac{1}{k} \mathbf{1}[a \in \mathcal{N}_k(v)], \quad (11)$$

$$\alpha_{v,a}^{\text{rand}} = \frac{z_a \mathbf{1}[a \in \mathcal{R}_k(v)]}{\sum_{a' \in \mathcal{R}_k(v)} |z_{a'}|}, \quad z_a \sim \mathcal{N}(0, 1), \quad (12)$$

where $\mathcal{N}_k(v)$ is the set of the k nearest candidates (the first k of the sorted $C(v)$) and $\mathcal{R}_k(v)$ is a set of k candidates drawn uniformly without replacement from $C(v)$, with one fixed realisation (transplant seed 0). MEAN asks whether the fitted sparse solve beats the crudest sensible weighting; RANDOM asks whether *any* placement in the anchor span, with the same sparsity, already delivers whatever benefit the pipeline has. Each transplant artefact is built once; downstream seeds vary fine-tuning only.

A measurement on the raw reconstructions showed that (10) inflates XLM-R rows: the median norm of reconstructed rows was $1.9091\times$ that of the base anchor rows (XLM-15: $1.1324\times$; Section VI-B). Because the pre-registered principle is that reconstructed rows must live at the base model’s own scale, every installed artefact, for both bases and all three initialisers, rescales each reconstructed row to the median anchor norm while preserving its direction:

$$\hat{\mathbf{b}}_v \leftarrow \hat{\mathbf{b}}_v \cdot \frac{\text{median}_{a \in \mathcal{A}} \|\mathbf{b}_a\|_2}{\|\hat{\mathbf{b}}_v\|_2}. \quad (13)$$

The six installed artefacts all report a reconstructed-to-anchor median-norm ratio of exactly 1.000 (Table V).

E. Installing the transplant

The donor tokenizer replaces the base tokenizer; the input-embedding table is resized to V_D and filled with the anchors’ own base rows for $a \in \mathcal{A}$, the aligned special-token rows, and $\hat{\mathbf{b}}_v$ for $v \in \mathcal{V}_{\text{unf}}$; the tied masked-LM output projection receives the same matrix and weights are re-tied. Nothing else in the checkpoint is modified. Algorithm 2 summarises the pipeline.

F. Fidelity controls

Before any training, each artefact passes controls that rule out serialisation and bookkeeping errors (Table V). The *identity* control maps the base vocabulary onto itself through the same anchor machinery, writes the embedding matrix, saves and reloads the model, and compares a forward pass: the embedding

TABLE IV

ANCHOR IDENTIFICATION. COUNTS FROM RESULTS/ANCHORS.JSON; “NAIVE” IS THE RAW STRING INTERSECTION BEFORE CANONICALISATION; SHARE IS FUNCTIONAL ANCHORS OVER THE DONOR VOCABULARY OF 32,770; “UNFAMILIAR” TOKENS RECEIVE RECONSTRUCTED EMBEDDINGS.

Base	Scheme	V_B	Naive	Strict	Surface	Functional	Share (%)	Unfamiliar
XLM-15	fastbpe	95,000	1,545	1,637	3,938	8,047	24.56	24,723
XLM-R	sentencepiece	250,002	2,856	10,662	9,539	9,776	29.83	22,994

Algorithm 2: Tokenizer transplantation (src/transplant/)

Input: E_B , E_D ; anchors $\mathcal{A}$; unfamiliar $\mathcal{V}_{\text{unf}}$; method $\in \{\text{omp}, \text{mean}, \text{random}\}$; $k=64$, $m=256$, seed 0
Output: transplanted checkpoint with donor tokenizer

```

1  $\tilde{D}_{\mathcal{A}} \leftarrow \text{row-normalise } E_D[\mathcal{A}]$ ;
2  $B_{\mathcal{A}} \leftarrow E_B[\mathcal{A}]$  (unnormalised);
3 foreach  $v \in \mathcal{V}_{\text{unf}}$  (batches of 256) do
4    $\tilde{d}_v \leftarrow E_D[v] / \|E_D[v]\|$ ;
5    $C(v) \leftarrow \text{top-}m \text{ anchors by } \tilde{d}_v^T \tilde{d}_a, \text{ sorted descending};$ 
6   if  $\text{method} = \text{omp}$  then
7      $\alpha \leftarrow \text{OMP}(\tilde{d}_v, \tilde{D}_{C(v)}, k)$ ; // (9)
8   else if  $\text{method} = \text{mean}$  then
9      $\alpha \leftarrow \frac{1}{k}$  on first  $k$  candidates; // (11)
10  else
11     $\alpha \leftarrow \text{random coefficients on } k \text{ drawn cand.}$ ; // (12)
12   $\hat{b}_v \leftarrow \alpha^T B_{C(v)}$ ; // coefficient transfer, (10)
13  $E' \leftarrow \text{resize to } V_D$ ;
14  $E'[a] \leftarrow E_B[u_a]$  for  $(a, u_a) \in \mathcal{A}$ ; align special rows;
15  $E'[v] \leftarrow \hat{b}_v$  for  $v \in \mathcal{V}_{\text{unf}}$ ;
16 rescale each  $E'[v]$  to  $\text{median}_a \|E'[a]\|$ ; // (13)
17 install  $E'$  as input table and tied output rows;
18 save with donor tokenizer;
19 run identity, round-trip, forward-pass controls (Sec. IV-F);
```

tables are bit-identical after the round trip, the maximum absolute logit difference is 0.0, and there are no canonical-key collisions. These checks establish that the transplant we evaluated is the transplant we built; they say nothing about whether reconstructed rows carry useful lexical information, which is the subject of the intrinsic diagnostics in Section VI-B.

G. Turkish intermediate stage and the word-shuffled control

Real Turkish training fine-tunes the encoder with a three-class sentiment head on 18,000 retained TRSAv1 examples [27] for two epochs, evaluating on 2,000 held-out Turkish examples per epoch; its product is the end-of-training checkpoint, built once per (base, tokenizer, corpus, seed)—30 distinct checkpoints, produced serially in a first phase—and reused read-only by the Azerbaijani stage. For OMP+Tr the Turkish stage runs on the transplanted model, so the Turkish stage itself is affected by the transplant. The shuffled control applies the identical procedure to a corpus in which the whitespace-delimited words of each example are independently permuted, keeping the example’s label and word multiset. The manipulation changed word order in 19,561 of 20,000 examples (97.805%) and preserved every multiset (results/scramble.json). Shuffling changes order, fluency, and source-task difficulty together, so real-minus-shuffled is an *order-sensitive* contrast, not a syntax-only intervention.

Source-stage quality is recorded, not assumed: every Turkish-bearing run stores its Turkish validation macro-F1 and a

degeneracy flag (macro-F1 < 0.40 or constant prediction over three classes). Degenerate stages are *reported*, never excluded, because excluding failed source stages would be an outcome-based exclusion.

H. Task-head reset

After the Turkish checkpoint is loaded, every module outside the encoder prefix is re-initialised with the model’s own initialiser, for every condition, so that no arm carries a differently aged head: 2,050 `sequence_summary` parameters for XLM-15 and 592,130 `classifier` parameters for XLM-R, recorded per run. This closes a defect found by a forensic audit of an earlier grid, in which `ignore_mismatched_sizes` re-initialised only the shape-mismatched output projection and silently carried the Turkish-trained dense projection into the Azerbaijani stage (Amendment 8.3). Consequently any Turkish-stage effect must pass through the encoder.

I. Two-stage training and checkpoint selection

Algorithm 3 gives the per-run procedure. All Azerbaijani runs use 2,000 optimizer steps regardless of n , so that update opportunity is held constant: an epoch budget would give roughly 63, 313, 1,250 and 6,250 steps across the size range and could prevent escape in the small cells by construction. Under the fixed budget the model sees the training subset about 125, 31.7, 6.4 and 3.1 times at $n = 500, 2,000, 10,000$ and 20,914 respectively; the size sweep is therefore a fixed-budget stress test, not a converged learning curve. Validation macro-F1 is recorded at every 200 steps; a checkpoint is saved at each evaluation point with `save_total_limit = 1`; the best-by-macro-F1 checkpoint is restored at the end and *re-scored on validation*, and the run aborts if the restored score differs from the recorded maximum by more than 10^{-4} . Only then is the fixed test split evaluated, once, with predictions stored and an entry appended to a persistent ledger. Early stopping is disabled, so every run trains all 2,000 steps and the collapse diagnostics see the full history.

J. Statistical analysis plan

All inference is within base and was fixed before the grid. For a cell, the escape count x of N seeds gets a Wilson 95% interval [33],

$$\frac{\hat{p} + \frac{z^2}{2N} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}, \quad z = 1.96, \quad (14)$$

and $\bar{F}_{\text{cond}}$, $\bar{F}_{\text{all}}$ get 10,000-resample percentile bootstrap intervals [35]. For a contrast between conditions A and C at the

TABLE V

FIDELITY CONTROLS PER BASE (RESULTS/CONTROLS___{BASE}.JSON, RESULTS/TRANSPLANT___{BASE}___*_RESCALED.JSON). RATIOS ARE RECONSTRUCTED-TO-ANCHOR MEDIAN NORMS OF THE INSTALLED OMP / MEAN / RANDOM ARTEFACTS AFTER EQ. (13).

Base	Identity matrix	Key collisions	Round trip bit-exact	Max $ \Delta \logit $	Logits identical	Norm ratios OMP / Mean / Random
XLM-15	yes	0	yes	0.0	yes	1.000 / 1.000 / 1.000
XLM-R	yes	0	yes	0.0	yes	1.000 / 1.000 / 1.000

Algorithm 3: One run (B, c, n, s) (src/training/finetune.py)
Input: base B , condition $c = (\kappa, \pi)$, size n , seed s ; frozen hyperparameters (Table IX)
1 set all seeds to s ; deterministic algorithms on;
2 load checkpoint for κ (original base, or transplanted artefact of B);
3 **if** $\pi \neq \text{none}$ **then**
4 **if** cache hit for (B, s, π) **then** load Turkish checkpoint;
5 **else** train 3-class head on 18k (shuffled) TR, 2 epochs; record TR macro-F1; cache;
6 reset every non-encoder module (task head); record what was reset;
7 sample n Azerbaijani training examples deterministically for seed s ;
8 **for** $t = 1$ **to** 2000 **do**
9 AdamW step: batch 32, lr $2 \cdot 10^{-5}$, warm-up 10%, wd 0.01, fp16;
10 **if** $t \bmod 200 = 0$ **then**
11 record $F(t)$ on validation; save checkpoint (keep best)
12 $E \leftarrow 1[\max_t F(t) > 0.40]$; $t^* \leftarrow$ earliest argmax ;
 // Eqs. (1), (2)
13 restore t^* ; re-score validation; **abort** if $|F_{\text{restored}} - F(t^*)| > 10^{-4}$;
14 evaluate test split once; store predictions; append ledger entry;
15 write schema-v4 record (history, selection, test, TR stage, reset, runtime);

Algorithm 4: Inference family (src/analysis/stats.py)
1 **foreach** base B , size n , unordered pair $\{A, C\}$ of conditions run at (B, n) **do**
2 Fisher exact test on escape counts (diagnostic);
3 **if** $\# \text{esc}(A) \geq 2$ **and** $\# \text{esc}(C) \geq 2$ **then**
4 Welch t on cond. macro-F1; bootstrap CI; Cohen's d ;
 // primary family
5 **else** mark NOT MEASURED;
6 **if** ≥ 2 shared escaped seeds **then** paired t (diagnostic);
7 all-seed paired t (sensitivity);
8 Holm-adjust the 86 Welch p ; survivors $\leftarrow \{p^{\text{Holm}} < 0.05\}$; // (16)
9 **foreach** survivor **do**
10 paired t on test macro-F1; Holm within survivors; exact sign-flip
 p ; // (17)

same base and size, escape rates are compared by Fisher's exact test (diagnostic). The *primary family* compares conditional macro-F1 by Welch's t [34],

$$t = \frac{\bar{F}_C - \bar{F}_A}{\sqrt{s_C^2/n_C + s_A^2/n_A}}, \quad v = \frac{(s_C^2/n_C + s_A^2/n_A)^2}{\frac{(s_C^2/n_C)^2}{n_C - 1} + \frac{(s_A^2/n_A)^2}{n_A - 1}}, \quad (15)$$

over escaped seeds, and is *measurable* only if both arms have at least two escaped seeds; 86 of the 96 declared contrasts are measurable. The family-wise error is controlled by Holm's step-down rule [32]: with ordered raw $p_{(1)} \leq \dots \leq p_{(M)}$, $M = 86$,

$$p_{(i)}^{\text{Holm}} = \max_{j \leq i} \min \{(M - j + 1) p_{(j)}, 1\}, \quad (16)$$

and a contrast *survives* if $p^{\text{Holm}} < 0.05$ (the plain Bonferroni threshold is $0.05/86 = 5.81 \times 10^{-4}$). Same-seed paired t tests on seeds escaped in both arms, all-seed paired tests, Cohen's d , and Fisher tests are reported as diagnostics or sensitivity analyses and are not members of the adjusted family. Test-split inference is restricted to *confirmation*: the 27 validation survivors are re-tested with a paired t on test scores, Holm-corrected within the 27; no new family is discovered on test. Finally, because five paired seeds admit only $2^5 = 32$ sign assignments, the exact two-sided sign-flip p -value [36],

$$p_{\text{exact}} = \frac{1}{32} \sum_{\sigma \in \{\pm 1\}^5} \mathbf{1}[|\bar{\delta}_\sigma| \geq |\bar{\delta}|] \geq \frac{2}{32} = 0.0625, \quad (17)$$

with $\bar{\delta}_\sigma = \frac{1}{5} \sum_i \sigma_i \delta_i$ the sign-flipped mean of the five paired seed differences δ_i and $\bar{\delta}$ the observed mean, can never reach 0.05; parametric survivors are therefore reported as model-based evidence, and (17) is computed for each of them as the distribution-free bound. Algorithm 4 summarises the family logic.

V. EXPERIMENTAL SETUP

A. Azerbaijani task data

The Azerbaijani source is the LocalDoc sentiment dataset on Hugging Face [26] (42,000 texts from social networks and reviews, three classes, licence CC-BY-NC-SA-4.0). The dataset exceeds the brief's floor of 10,000 labelled examples; the primary cells at $n \in \{500, 2,000\}$ are deliberate low-resource subsamples of a fixed training split, chosen because that is where transfer effects are expected and where escape instability was measured in pilots, and the full-size cell $n = 20,914$ establishes the ceiling. Table VI traces the pipeline. Neutral examples are removed to obtain a binary task; label-aware exact-text deduplication removes 86 duplicates (0.31%) with zero label conflicts, *before* a stratified split with seed 12,345. The resulting splits are balanced, and an exact-text leakage check finds no overlap between any pair of splits. Because the upstream Hugging Face revision is not pinned, an order-invariant content manifest (SHA-256 over sorted normalised text-label pairs) is recorded for every split; the supplied data.tgz snapshot is the only exact copy of the data the grid saw, and any reproduction must verify the manifests before treating a fresh download as equivalent. No scraping was performed and no personal data beyond the public dataset was used.

TABLE VI

AZERBAIJANI DATA PIPELINE AND SPLITS (RESULTS/SPLITS.JSON). THE PRE-GRID CHARACTERISATION PACKAGE (SECTION V-B) USED 20,936/2,791/4,187 AND CARRIES NO MANIFEST.

Step	Count
Upstream rows (positive / neutral / negative)	42,000
After removing <code>neutral</code>	28,000
After label-aware exact-text deduplication (86 removed, 0 conflicts)	27,914
Train / validation / test (stratified, seed 12,345)	20,937 / 2,791 / 4,186
Train label distribution (negative / positive)	10,472 / 10,465
Test label distribution (negative / positive)	2,094 / 2,092
Exact-text leakage train-test / train-val / val-test	0 / 0 / 0
Content manifest, test split (SHA-256 prefix)	ca19c0bb
Content manifest, validation / train (prefix)	5cd479fb / d1cf165c
Upstream revision pinned	no

TABLE VII

HUMAN LABEL AUDIT, 100 SAMPLED EXAMPLES
(EVIDENCE/CARRIED_FORWARD_PRE_GRID/).

Quantity	Value
Three-class human-human agreement	66/99 = 66.67% ($\kappa = 0.491$)
Human-human on retained polar rows, unconditioned	44/67 = 65.67%
Human-human, all three sources commit to a polarity	37/42 = 88.10% ($\kappa = 0.746$)
Annotator 1 vs. gold (100 / retained 67)	71.0% / 68.66%
Annotator 2 vs. gold (100 / retained 67)	58.59% / 59.70%
Audit verdict	PROCEED_WITH_CAUTION

B. Human label audit (pre-grid)

Two annotators independently labelled 100 examples sampled under the pre-grid split (Table VII). Three-class human-human agreement is 66/99 (66.67%, $\kappa = 0.491$). Among rows retained by the binary task, the *unconditioned* agreement is 44/67 (65.67%); the higher 37/42 (88.10%, $\kappa = 0.746$) conditions on both humans and the dataset all committing to a polarity and therefore selects easier examples. The unconditioned figure is the relevant disclosure of retained-label quality, and the two annotators are not interchangeable (68.66% vs. 59.70% against gold). Because the pre-grid split differs from the grid split by one example and has no manifest, these numbers characterise the dataset, not the specific holdout.

C. Turkish source data

TRSAv1 [27] provides 150,000 Turkish e-commerce reviews with three labels; the captured dataset card declares no licence. Label-aware deduplication removes 11,360 rows (7.57%; 937 texts had conflicting labels), leaving 138,640, from which a seeded, label-stratified subsample of 20,000 is deterministically partitioned into 18,000 source-training and 2,000 source-validation examples. The shuffled corpus is derived from the same 20,000 (Section IV-G).

D. Pre-registration and amendments

The experimental specification was frozen on 4 September 2026 and hash-locked (`configs/CONFIG_HASHES.lock`); every later change is an amendment recorded with its date and rationale before the affected runs (Table VIII). Two classes of amendment exist. Non-scientific execution changes

TABLE VIII

PRE-REGISTRATION TIMELINE
(EVIDENCE/PREREGISTRATION/FROZEN.MD).

Date (2026)	Event
04 Sep	Specification frozen; hash lock; $\theta = 0.40$ fixed from pilots
04 Sep	Validation cadence $65 \rightarrow 200$ steps; eval batch size; A100/H100 migration notes; gradient accumulation 8×4 authorised
05 Sep	Accumulation reverted to pre-registered batch 32×1 (no run had completed under 8×4)
06 Sep	114-run validation-only grid (retained separately, not pooled)
07 Sep	Amendments 1–5: Turkish conditions and contrast base at $n = 500$; test split spent once with ledger; schema v4; per-method C1c provenance
07 Sep	Amendment 6: best-validation checkpoint restore with 10^{-4} verification
08 Sep	Amendment 8: nearest- k mean placebo; UNK/special rejection in anchors; full task-head reset; degeneracy accounting; all-seed estimand, paired tests, Holm; manifests; valid JSON
08–09 Sep	Final 164-run grid executed (19:39 UTC 8 Sep to 03:58 UTC 9 Sep)

(validation cadence $65 \rightarrow 200$ steps, evaluation batch size, two-phase execution, a gradient-accumulation option adopted and then reverted before any run completed) do not alter what the optimizer does at any step. Amendments 6 and 8, written before the final grid, repair defects found by a forensic audit of an earlier 114-run grid: validation-selected checkpoint restoration with numerical verification, a genuinely reset task head, a well-defined mean placebo, UNK-free anchors, degeneracy accounting, dual estimands with family-wise correction, and content manifests. That earlier grid is retained unmodified as a separate artefact and is not pooled with the 164 runs reported here. The escape threshold $\theta = 0.40$, the seeds, sizes, k , m , and every optimizer setting were never changed after the freeze.

E. Grid, seeds, and hyperparameters

Table IX lists the frozen values. At $n = 500$ and $n = 2,000$ both encoders run all seven conditions with seeds {42, 1337, 13, 7, 2024} (140 runs); XLM-15 additionally runs the four non-Turkish conditions at $n \in \{10,000, 20,914\}$ with seeds {42, 1337, 13} (24 runs). The launcher records 164 requested, 164 completed, zero failures, and no skipped-existing records. The queue is ordered reference-size-first so that a truncated window would still yield a complete comparison table.

F. Test-split protocol and disclosures

The inference family is defined on validation. The test split is evaluated once per run, only at the restored validation-selected checkpoint, and only after the restore has been numerically verified; the ledger `results/test_evaluation_ledger.jsonl` holds exactly 164 entries, one per completed run, all from the single call site `finetune.run_single`, which defaults to the validation split and evaluates the test split only with an explicit opt-in flag. No model, step, hyperparameter, or condition was selected using a test number; test-side statements in Section VI are confirmations of validation-selected claims (Section IV-J).

G. Compute environment and budget

The 164-run grid executed on a rented instance with one NVIDIA H100 NVL (95,830 MiB), an AMD EPYC 9V84

TABLE IX
FROZEN HYPERPARAMETERS AND GRID COMPOSITION
(CONFIGS/EXPERIMENT.YAML, FROZEN.MD).

Setting	Value
Azerbaijani budget	2,000 optimizer steps (all n)
Batch / accumulation / effective batch	32 / 1 / 32
Optimizer	AdamW [37], lr 2×10^{-5} , warm-up 10%, weight decay 0.01
Precision / max length / padding	fp16 AMP / 256 / dynamic
Validation cadence	every 200 steps (10 points incl. terminal)
Selection rule	max validation macro-F1, ties $\rightarrow$ lower step; restore and verify (10^{-4})
Early stopping	disabled
Escape threshold θ	0.40 (fixed from pilots, hash-locked)
Turkish stage	2 epochs, 18k train / 2k val, 3 labels, cached (30 checkpoints)
Transplant	functional anchors; $k = 64$; $m = 256$; rescale to anchor median; transplant seed 0
Seeds ($n \leq 2,000$ / $n > 2,000$)	{42, 1337, 13, 7, 2024} / {42, 1337, 13}
Split seed	12,345
Runs: $2 \times 7 \times 2 \times 5 + 1 \times 4 \times 2 \times 3$	140 + 24 = 164 (164 completed, 0 failed)
Determinism	use_deterministic_algorithms; cuDNN deterministic

TABLE X
COMPUTE COST OF THE FINAL GRID
(EVIDENCE/ENVIRONMENT/ENVIRONMENT_AS_RUN.MD, RESULTS/AGGREGATE.CSV, RESULTS/LAUNCHER_STATE.JSON). PER-RUN WALL-CLOCK IN MINUTES.

Base	Runs	Process-h	Mean	Min	Max
XLM-15	94	5.00	3.19	2.94	3.60
XLM-R	70	3.36	2.88	2.53	6.77
All	164	8.36	3.06	2.53	6.77

Azerbaijani-stage process-hours: 8.02; launcher elapsed: 8.37 h; streams: 1. Wall-clock incl. the serial Turkish-cache phase ≈ 9.5 h $\Rightarrow \approx 9.5$ GPU-hours on one GPU. Peak GPU memory during the grid: NOT MEASURED (sampler did not run). Pre-grid 30-step benchmark on an RTX 4060 at the same batch and length: 8.6 GB peak reserved.

(40 vCPU), 313 GiB RAM, driver 595.71 with CUDA 13.2 host and PyTorch 2.11.0+cu128, in a single stream throughout (streams_active=1 in all 164 records). Table X summarises cost: 8.36 process-hours summed over runs, of which 8.02 h is the Azerbaijani stage; launcher wall-clock 8.37 h; roughly 9.5 h including the serial Turkish-cache phase; hence about 9.5 device-hours. Per-run wall-clock is 2.5–6.8 minutes (Fig. 2). This is far inside the brief’s two-day training envelope. Peak GPU memory during the grid is NOT MEASURED: the launcher’s memory sampler did not run, no run record carries a maximum allocation, and the 97 MiB reading captured after completion describes an idle device. A 30-step benchmark at the locked batch and sequence settings on a different, pre-grid machine (RTX 4060 laptop, 8.59 GB) measured 8.6 GB peak reserved memory; we report it as a labelled pre-grid measurement on other hardware, not as the grid’s footprint, and note that a stream ceiling of 10 GB was enforced by the launcher configuration on every venue.

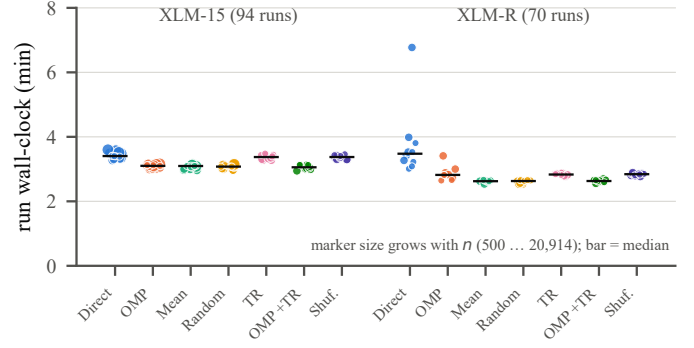


Fig. 2. Per-run wall-clock by base and condition, all 164 runs (results/aggregate.csv). Marker size grows with n ; the bar is the cell median. Turkish-bearing runs exclude the cached source stage.

TABLE XI
PRE-RUN PINS VS. THE STACK THAT ACTUALLY RAN
(EVIDENCE/ENVIRONMENT/).

Package	Pre-run lock	As run
torch	2.4.1+cu121	2.11.0+cu128
transformers	4.44.2	4.57.6
tokenizers	0.19.1	0.22.2
datasets	2.21.0	5.0.1
numpy	1.26.4	2.5.3
scipy	1.14.1	1.18.1
scikit-learn	1.5.2	1.9.0
pandas	2.2.3	3.0.5
matplotlib	3.9.2	3.11.1
sentencepiece	0.2.0	0.2.2
sacremoses	0.2.0	0.2.0 (undeclared)
protobuf	absent	7.36.1 (undeclared)

H. Software stack and reproducibility

The executed environment used Python 3.12.14, PyTorch 2.11.0+cu128 [38], Transformers 4.57.6 [39], Tokenizers 0.22.2, Datasets 5.0.1, NumPy 2.5.3, SciPy 1.18.1, scikit-learn 1.9.0 [40], pandas 3.0.5 and Matplotlib 3.11.1 (Table XI). Two disclosures follow from the environment capture. The pre-run requirements.lock was unresolvable on the host because of a Datasets/fsspec conflict, so it did not describe the run; sacremoses (needed by the XLM-15 tokenizer) and protobuf (needed for SentencePiece conversion) were undeclared and installed after import failures. requirements.txt and requirements.lock were regenerated from the post-run freeze and describe the environment the reported numbers were computed on; they do not retroactively prove that the original specification was sufficient.

All seeds, sizes, paths and analysis rules are read from configs/experiment.yaml; run_all.sh is the single entry point that rebuilds anchors, transplants, controls, the Turkish cache, the 164-run grid, and the aggregate, statistics, and table files in order, with automatic resume through skip_existing. The repository’s continuous-integration workflow (.github/workflows/ci.yml) runs the CPU unit-test suite (pytest -m ‘not slow’, offline, no model or dataset downloads) together with repository-hygiene checks on every push to and pull request against main. Every result file

TABLE XII

TOKENIZER DIAGNOSTICS. FERTILITY, ALPHABETIC TYPES AND OVERLAP FROM RESULTS/OVERLAP_CONTROL.JSON (5,000 WIKIPEDIA SENTENCES PER LANGUAGE); LETTER COUNTS ARE THE NUMBER OF VOCABULARY ENTRIES CONTAINING THE LETTER (EVIDENCE/CARRIED_FORWARD_PRE_GRID/DIACRITICS.JSON, PRE-GRID, DATASET-LEVEL); TOKEN-LENGTH STATISTICS AND TRUNCATION FROM RESULTS/TRUNCATION.JSON ON THE 27,914 TASK EXAMPLES. LIFT = TURKISH OVERLAP MINUS THE ENGLISH-FINNISH MEAN.

Tokenizer	Vocab	Fertility	AZ alpha.	AZ types covered by (%)					Vocab. entries with			Task tokens		Truncated (%)	
			types	TR	EN	FI	RU	Lift	ə	ş	ğ	mean	p99	128	256
XLM-15	95,000	3.206	3,257	66.7	57.4	59.0	16.8	+8.5	2	0	0	49.0	283	5.16	1.26
XLM-R	250,002	1.904	10,054	46.4	34.1	33.6	9.0	+12.6	1,366	891	407	33.7	189	2.31	0.47
AZ donor	32,770	1.758	14,858	32.1	19.3	19.1	7.7	+12.9	8,814	2,253	928	30.3	169	1.81	0.35

cited in this paper is committed: 164 schema-v4 run records (each with the ten-point validation history, the training-loss history, the test block with all 4,186 predictions, the selection and restoration fields, and the task-head-reset and source-stage metadata), the per-run logs, `aggregate.csv`, `summary.csv`, `stats.csv`, the intrinsic-diagnostic and control files, the environment captures, and the pre-registration documents (Appendix E). Per-run model weights and the 30 Turkish-stage checkpoints are not distributed (tens of GB); their reconstruction costs roughly 2.5 GPU-hours.

VI. RESULTS

All numbers below are read from the committed result files; the figure and table scripts in `report/ieee/` regenerate every artefact from those files. Values are reported to the precision of the source.

A. Tokenizer-level diagnostics

Fig. 3 and Table XII characterise the three tokenizers on 5,000 Azerbaijani Wikipedia sentences (102,726 words) and on the 27,914 task examples. XLM-15 segments Azerbaijani into 3.21 tokens per word, XLM-R into 1.90, and the donor into 1.76; the XLM-15 deficit is structural, because its lower-cased, accent-stripped fastBPE vocabulary contains no token bearing ş, ç, ö, ü or ğ and only two containing ə, whereas XLM-R and the donor contain thousands (pre-grid vocabulary counts, `diacritics.json`). At the frozen maximum length of 256 tokens, 1.257% of task examples are truncated under XLM-15, 0.473% under XLM-R, and 0.351% under the donor; the transplant therefore also changes how much text the encoder sees, a confound of at most 0.9 percentage points of examples.

Alphabetic token-type overlap between Azerbaijani and four other languages (Fig. 3b) shows that Turkish covers more Azerbaijani types than English or Finnish for every vocabulary—66.7% vs. a 58.2% Latin-script baseline for XLM-15 (+8.5 pt), 46.4% vs. 33.8% for XLM-R (+12.6 pt), 32.1% vs. 19.2% for the donor (+12.9 pt)—while Cyrillic-script Russian covers 8–17%. The English–Finnish spread is at most 1.6 pt, so the Turkish lift is not noise, but the Russian comparison shows that script alone moves overlap by tens of points; overlap is a lexical-similarity signal, not evidence of linguistic transfer, and it is the reason a lexical-only explanation of any Turkish benefit cannot be excluded by design.

B. Intrinsic transplant diagnostics

Before any downstream training, each transplanted artefact was probed on a deterministic 300-line sample of Azerbaijani *training* text (validation and test untouched) with three intrinsic measures (Table XIII, Fig. 4): masked-token top-1 accuracy of the masked-LM head, bits per character (BPC), and the embedding-norm ratio before rescaling. BPC rather than perplexity is used because different tokenizers split the same text into different numbers of tokens; BPC normalises the masked-LM loss by characters:

$$\text{BPC} = \frac{\ell_{\text{MLM}} \cdot N_{\text{tok}}}{N_{\text{char}} \ln 2}, \quad (18)$$

with ℓ_{MLM} the mean loss per masked token in nats.

The two bases behave very differently. XLM-R’s original masked-LM head predicts 602 of 1,631 masked Azerbaijani tokens (36.91%) at 1.60 BPC; after OMP transplantation it predicts 40 of 1,502 (2.66%) at 6.94 BPC ($\Delta = +5.34$), and the raw reconstruction inflated row norms by 1.91× before rescaling, outside the healthy band of [0.85, 1.15] defined in the norm report. The placebos are worse still ($\Delta\text{BPC} +5.53$ for MEAN, +7.06 for RANDOM), so OMP is the *best* of the three initialisers on XLM-R while remaining far from the original. XLM-15’s original head is already near chance on Azerbaijani (2 of 2,323, 0.09%; 7.21 BPC), and its OMP transplant stays at 2 of 1,502 with a *lower* BPC of 6.00 ($\Delta = -1.20$) and a healthy norm ratio of 1.13. As the quality file itself warns, a BPC decrease at near-zero top-1 accuracy is a calibration change over a smaller vocabulary, not recovered lexical knowledge; the two measures must be read together. Eligible masked positions differ between tokenizers, so these are diagnostics rather than paired token-level accuracies. The independent signals—top-1 collapse, BPC increase, and norm inflation—all point to the same conclusion the downstream grid reaches for XLM-R: the implemented reconstruction does not preserve the information the encoder expects in its embedding rows.

C. Reference-size grid

Tables XIV and XV report every cell at the two primary sizes on validation and on test; Appendix A adds the confidence intervals and the higher-data cells. Two facts organise everything that follows. First, XLM-R escapes in every one of its 70 runs, so its conditional and all-seed estimands coincide and every XLM-R contrast is measurable. Second, XLM-15

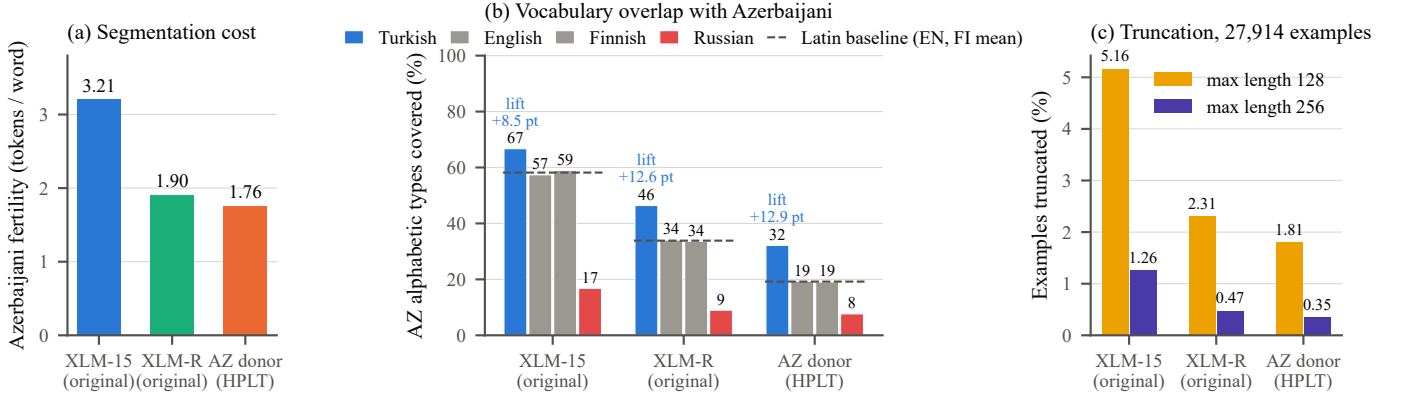


Fig. 3. Tokenizer diagnostics. (a) Azerbaijani fertility on 5,000 Wikipedia sentences (`results/overlap_control.json`). (b) Share of Azerbaijani alphabetic token types also present in the tokenization of 5,000 sentences of Turkish, English, Finnish and Russian; the dashed line is the English–Finnish mean, the annotated lift is Turkish minus that baseline. (c) Share of the 27,914 task examples truncated at maximum length 128 and 256 (`results/truncation.json`); 256 is the frozen setting.

TABLE XIII

INTRINSIC DIAGNOSTICS OF THE INSTALLED ARTEFACTS ON 300 AZERBAIJANI TRAINING LINES (`RESULTS/CONTROLS_{BASE}_{TAG}.JSON`). TOP-1 IS CORRECT/MASKED POSITIONS; BPC PER EQ. (18); THE NORM RATIO IS RECONSTRUCTED-TO-ANCHOR MEDIAN NORM OF THE RAW OMP RECONSTRUCTION BEFORE EQ. (13) (`EMBEDDING_NORM_REPORT`); ALL INSTALLED ARTEFACTS ARE RESCALED TO 1.000.

Base	Init.	Top-1 (correct/masked)		BPC		Pre-rescale	
		original	transplanted	original	transpl.	Δ	norm ratio
XLM-15	OMP	2/2,323	2/1,502	7.2071	6.0034	-1.2037	1.1324
XLM-15	Mean	2/2,323	0/1,502	7.2071	6.0763	-1.1308	–
XLM-15	Random	2/2,323	6/1,502	7.2071	8.1814	+0.9743	–
XLM-R	OMP	602/1,631	40/1,502	1.5975	6.9352	+5.3377	1.9091
XLM-R	Mean	602/1,631	19/1,502	1.5975	7.1276	+5.5301	–
XLM-R	Random	602/1,631	16/1,502	1.5975	8.6611	+7.0636	–

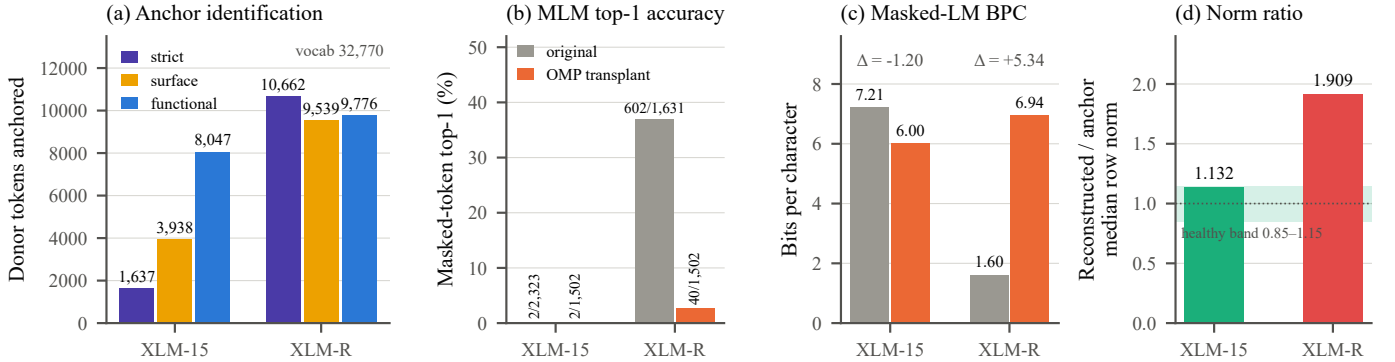


Fig. 4. Intrinsic transplant diagnostics. (a) Anchors found by the three identification modes (`results/anchors.json`); functional mode is the pre-registered default. (b) Masked-token top-1 accuracy of the original and the OMP-transplanted masked-LM head on the same 300 training lines (`results/top1_accuracy.json`). (c) Bits per character (`results/cross_base_transplant_quality.json`). (d) Reconstructed-to-anchor median-norm ratio of the raw OMP reconstruction with the healthy band of the norm report (`results/embedding_norm_report_*.json`); the installed artefacts were rescaled to 1.000.

escapes in 60 of 94 runs, between 2/5 and 5/5 per cell at the primary sizes, so its two estimands diverge by up to 0.21 and its conditional means describe different survivor subsets in different conditions.

For XLM-R at $n = 2,000$, OMP minus DIRECT is -0.0659 on validation ($0.8277 \rightarrow 0.7618$) and -0.0566 on test ($0.7970 \rightarrow 0.7404$); at $n = 500$ the effects are -0.0782 and

-0.0748 . MEAN (-0.0505 , -0.0560) and RANDOM (-0.0980 , -0.0996) also underperform DIRECT at both sizes, with RANDOM worst, and OMP sits between them; better than RANDOM by $+0.032$ and $+0.021$, worse than MEAN by -0.016 and -0.022 . TR is the numerically best XLM-R condition at both sizes (0.8307 and 0.8149 on validation; 0.8048 and 0.7910 on test), followed by SHUF, TR and DIRECT; OMP+TR lies between OMP

TABLE XIV

COMPLETE RESULTS AT $n = 2,000$. ESCAPE IS ESCAPED/DECLARED SEEDS; CONDITIONAL MACRO-F1 AVERAGES ESCAPED SEEDS ONLY (10,000-RESAMPLE PERCENTILE BOOTSTRAP 95% CI); ALL-SEED MACRO-F1 CHARGES COLLAPSED SEEDS AT THEIR ACTUAL SCORE; TEST VALUES ARE FROM THE RESTORED VALIDATION-SELECTED CHECKPOINT; TR DEG. COUNTS DEGENERATE TURKISH SOURCE STAGES (RESULTS/SUMMARY.CSV, T1 OF EVIDENCE/SUPPLEMENTAL/TEST_TABLES_RECEIVED.MD).

Base	Condition	Escape	Validation macro-F1			Test macro-F1		TR deg.
			cond.	95% CI	all-seed	cond.	all-seed	
XLM-15	Direct	2/5	0.6791	[0.674, 0.684]	0.4799	0.6735	0.4790	0
XLM-15	OMP	3/5	0.6148	[0.464, 0.691]	0.5023	0.6076	0.4980	0
XLM-15	Mean	3/5	0.6955	[0.675, 0.706]	0.5507	0.6833	0.5434	0
XLM-15	Random	4/5	0.6699	[0.664, 0.673]	0.6026	0.6529	0.5890	0
XLM-15	TR	4/5	0.6747	[0.661, 0.687]	0.6064	0.6663	0.5997	2
XLM-15	OMP+TR	5/5	0.6712	[0.645, 0.689]	0.6712	0.6645	0.6645	3
XLM-15	Shuf. TR	2/5	0.6873	[0.682, 0.692]	0.4750	0.6784	0.4714	2
XLM-R	Direct	5/5	0.8277	[0.823, 0.833]	0.8277	0.7970	0.7970	0
XLM-R	OMP	5/5	0.7618	[0.755, 0.769]	0.7618	0.7404	0.7404	0
XLM-R	Mean	5/5	0.7772	[0.772, 0.783]	0.7772	0.7578	0.7578	0
XLM-R	Random	5/5	0.7297	[0.725, 0.735]	0.7297	0.7154	0.7154	0
XLM-R	TR	5/5	0.8307	[0.829, 0.832]	0.8307	0.8048	0.8048	0
XLM-R	OMP+TR	5/5	0.7708	[0.766, 0.776]	0.7708	0.7546	0.7546	0
XLM-R	Shuf. TR	5/5	0.8269	[0.825, 0.829]	0.8269	0.7971	0.7971	0

TABLE XV

COMPLETE RESULTS AT $n = 500$; COLUMNS AS IN TABLE XIV.

Base	Condition	Escape	Validation macro-F1			Test macro-F1		TR deg.
			cond.	95% CI	all-seed	cond.	all-seed	
XLM-15	Direct	3/5	0.6306	[0.621, 0.639]	0.5117	0.6229	0.5071	0
XLM-15	OMP	2/5	0.5272	[0.440, 0.614]	0.4109	0.5248	0.4100	0
XLM-15	Mean	2/5	0.6359	[0.627, 0.645]	0.4544	0.6302	0.4521	0
XLM-15	Random	3/5	0.6192	[0.609, 0.632]	0.5049	0.6025	0.4949	0
XLM-15	TR	4/5	0.6141	[0.575, 0.639]	0.5580	0.6090	0.5539	2
XLM-15	OMP+TR	5/5	0.6217	[0.611, 0.630]	0.6217	0.6142	0.6142	3
XLM-15	Shuf. TR	3/5	0.6348	[0.620, 0.646]	0.5143	0.6336	0.5135	2
XLM-R	Direct	5/5	0.7980	[0.794, 0.804]	0.7980	0.7783	0.7783	0
XLM-R	OMP	5/5	0.7198	[0.710, 0.728]	0.7198	0.7035	0.7035	0
XLM-R	Mean	5/5	0.7421	[0.732, 0.752]	0.7421	0.7217	0.7217	0
XLM-R	Random	5/5	0.6985	[0.688, 0.709]	0.6985	0.6867	0.6867	0
XLM-R	TR	5/5	0.8149	[0.810, 0.819]	0.8149	0.7910	0.7910	0
XLM-R	OMP+TR	5/5	0.7261	[0.708, 0.741]	0.7261	0.7173	0.7173	0
XLM-R	Shuf. TR	5/5	0.8054	[0.799, 0.810]	0.8054	0.7814	0.7814	0

and MEAN. For XLM-15, conditional means at $n = 2,000$ lie within 0.61–0.70 for every condition while all-seed means range from 0.48 (DIRECT, 2/5 escaped) to 0.67 (OMP+TR, 5/5); the ordering of conditions depends on which estimand is chosen.

D. Optimisation failure: escape, collapse, and dynamics

Fig. 5 shows the ten validation points of all 70 runs at $n = 2,000$. Every XLM-R run leaves the collapse basin by the first or second evaluation (68 of 70 at step 200, 2 at step 400) and then changes little, which is why its seeds cluster tightly. XLM-15 behaves qualitatively differently: 34 of its 94 runs never leave the basin, escapes are spread from step 200 to step 2,000 with a mode at 400 (Fig. 6a), and individual runs can leave the basin, fall back into it, and leave again (TR, seed 42: 0.42 at step 600, 1/3 at steps 800–1,600, 0.65 at step 2,000). Two runs crossed the threshold mid-training and ended collapsed (OMP and MEAN at $n = 500$); they

are legitimately “escaped” under Eq. (1) while their selected checkpoint is an early one, and the schema records them as such (escaped_but_terminal_collapsed). Collapse is a one-class output: 32 of the 34 collapsed runs predict the negative class for every one of the 4,186 test examples and the remaining two predict it for almost all (Fig. 6b), giving test macro-F1 of 0.333–0.387.

No escape-rate difference between conditions is significant: across all 96 declared contrasts the smallest Fisher exact p is 0.167 (DIRECT 2/5 vs. OMP+TR 5/5 on XLM-15 at $n = 2,000$). With five seeds per cell, escape probabilities cannot be resolved beyond the Wilson intervals shown in Fig. 7a; this is a design limit stated in advance, not a post hoc finding.

E. Direct-relative effects and multiplicity control

Table XVI and Fig. 8 report every within-base contrast for XLM-R, where all seeds escaped and the conditional Welch

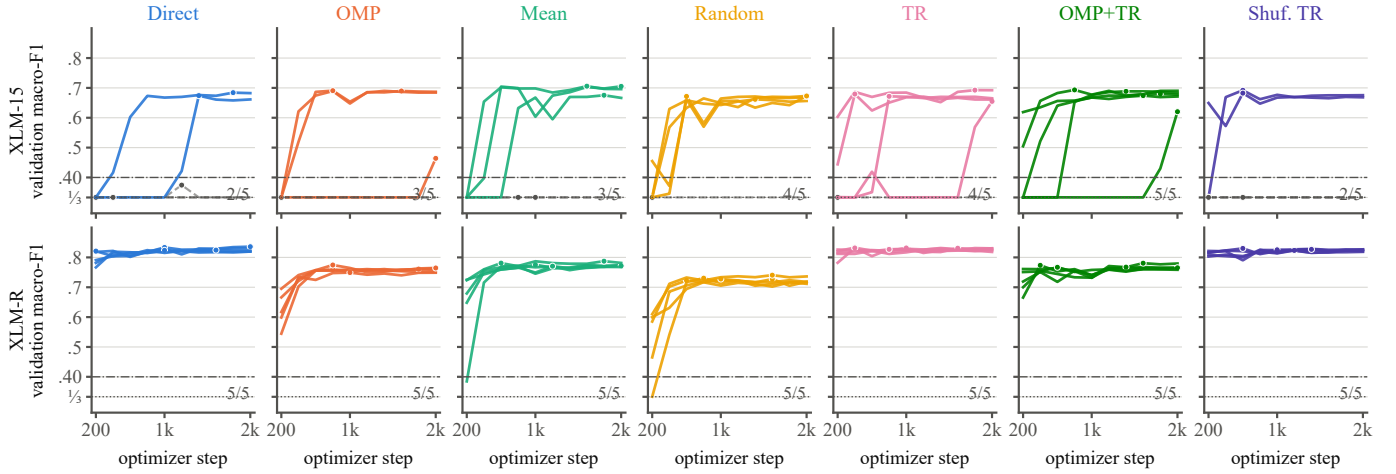


Fig. 5. Validation macro-F1 at the ten evaluation points (steps 200–2,000) of all 70 runs at $n = 2,000$, one panel per base and condition, one line per seed (results/runs/*_n=2000*.json, step_history). Coloured solid lines escaped; grey dashed lines never crossed the threshold. The dot marks the selected checkpoint t^* ; the dotted line is the collapse basin ($1/3$), the dash-dotted line the escape threshold (0.40); the corner number is escaped/declared seeds.

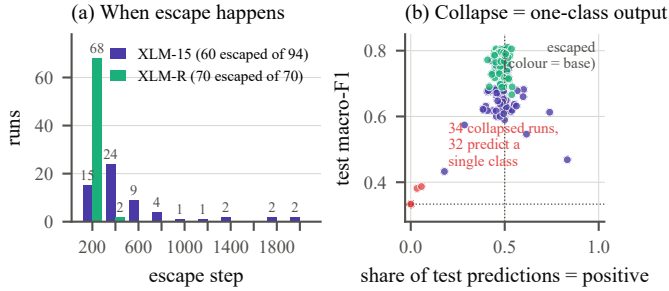


Fig. 6. Escape and collapse over all 164 runs (results/runs/*.json). (a) Step at which validation macro-F1 first exceeded 0.40 for the 130 escaped runs. (b) Share of positive predictions on the test split against test macro-F1; collapsed runs (red) sit at a single-class output with macro-F1 $\approx 1/3$.

family is fully measurable. Of the 86 measurable contrasts in the pre-registered family (all 42 XLM-R contrasts and 44 of the 54 XLM-15 contrasts), 27 survive Holm correction; every survivor is an XLM-R contrast (12 at $n = 500$, 15 at $n = 2,000$), and none involves XLM-15, whose smallest raw Welch p is 0.107. The survivors are the contrasts that separate the four donor-tokenizer arms (OMP, MEAN, RANDOM, OMP+TR) from the three original-tokenizer arms (DIRECT, TR, SHUF. TR)—all twelve such pairs at $n = 2,000$ and eleven of twelve at $n = 500$, where $\text{DIRECT} \rightarrow \text{OMP+TR}$ misses at Holm $p = 0.063$ —plus the within-transplant orderings that place RANDOM below MEAN at both sizes and below OMP and OMP+TR at $n = 2,000$. OMP versus MEAN does not survive at either size. Every one of the six Direct-relative transplant contrasts (three initialisers at two sizes) survives, with Cohen’s d between -5.3 and -14.9 and bootstrap intervals excluding zero; the Holm-adjusted p for $\text{DIRECT} \rightarrow \text{OMP}$ is 7.6×10^{-4} at $n = 500$ and below 10^{-5} at $n = 2,000$.

Turkish intermediate training is not a resolved Direct-relative improvement. TR minus DIRECT on XLM-R is $+0.0169$ at

$n = 500$ (Welch $p = 0.0034$, Holm-adjusted $p = 0.1955$; paired diagnostic $p = 0.0021$) and $+0.0030$ at $n = 2,000$ (Welch $p = 0.372$); the corresponding test differences are $+0.0127$ and $+0.0078$. The $n = 500$ direction is 5.6 times the $n = 2,000$ direction, which is compatible with the low-resource pattern reported by Kardeş-NLU, but the current seed count does not resolve it after correction for the family it belongs to. Real versus shuffled Turkish differs by $+0.0096$ at $n = 500$ (Welch $p = 0.048$, Holm 1) and $+0.0038$ at $n = 2,000$ (Welch $p = 0.038$, Holm 1): the order-sensitive component (H1) is directionally positive and small, and not established. SHUF. TR is itself indistinguishable from DIRECT ($+0.0073$ and -0.0008). The interaction contrasts are informative in one direction only: OMP+TR is below TR by -0.0889 and -0.0599 (both survive), showing that the transplant offsets the stronger original-tokenizer outcome; OMP+TR versus OMP ($+0.0062$, $+0.0090$; Holm 1) does not show that Turkish knowledge failed or succeeded in transferring through donor rows.

F. Test-side confirmation of validation-selected contrasts

The 27 survivors were fixed by the validation analysis and then re-tested on the held-out split with a paired t over the same five seeds, Holm-corrected within the 27 (Appendix B). All 27 retain their sign and remain significant on test (largest within-family Holm-adjusted p : 1.72×10^{-2} , for $\text{MEAN} \rightarrow \text{RANDOM}$ at $n = 500$); the mean test-to-validation absolute-effect ratio is 0.874, i.e., effects shrink slightly, as expected for a checkpoint selected on validation. Across the broader set of 38 paired contrasts (both bases, any shared escaped seed), validation and test differences agree in sign for 33, and for 26 of the 27 with $|\Delta_{\text{val}}| \geq 0.01$; the Pearson correlation between Δ_{val} and Δ_{test} is 0.9898 (Fig. 11c). Every sign disagreement occurs where $|\Delta_{\text{val}}| < 0.01$ or where fewer than three seeds escaped in both arms. These calculations assess stability of the validation-

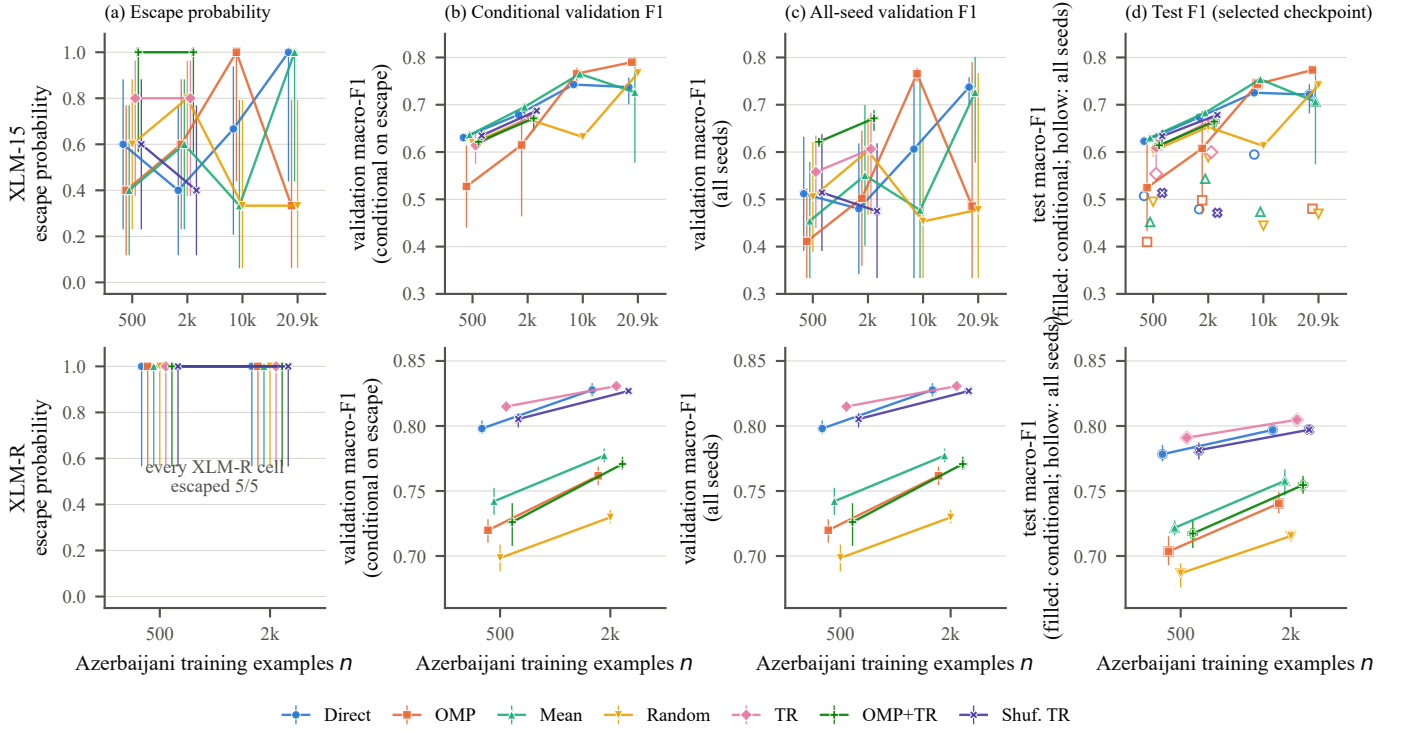


Fig. 7. Failure-aware summary across Azerbaijani training sizes (top: XLM-15; bottom: XLM-R). (a) Escape probability with Wilson 95% intervals; every XLM-R cell escaped 5/5. (b) Validation macro-F1 conditional on escape and (c) over all seeds, with 10,000-resample percentile-bootstrap intervals (results/summary.csv). (d) Test macro-F1 at the validation-selected checkpoint, conditional (filled, with CI) and all-seed (hollow) (T1 of test_tables_received.md). Lines connect matched conditions and are visual guides; the size axis is a fixed-budget stress test, not a learning curve.

TABLE XVI

DIRECT-RELATIVE CONTRASTS ON XLM-R (RESULTS/STATS.CSV). Δ IS CONDITIONAL VALIDATION MACRO-F1 (COMPARISON MINUS DIRECT) WITH ITS BOOTSTRAP 95% CI; WELCH p IS RAW, HOLM p IS ADJUSTED OVER THE 86-TEST FAMILY; PAIRED p IS THE SAME-SEED DIAGNOSTIC; Δ TEST IS THE PAIRED TEST DIFFERENCE (T2 OF TEST_TABLES_RECEIVED.MD); ✓ MARKS FAMILY SURVIVORS.

n	Contrast	Δ val	95% CI	Welch p	Holm p	Paired p	d	Δ test	Surv.
500	Direct $\rightarrow$ OMP	-0.0782	[-0.090, -0.068]	1.0e-5	7.6e-4	3.5e-4	-7.99	-0.0748	✓
500	Direct $\rightarrow$ Mean	-0.0560	[-0.068, -0.044]	1.4e-4	0.0090	0.0026	-5.31	-0.0566	✓
500	Direct $\rightarrow$ Random	-0.0996	[-0.112, -0.088]	1.0e-5	7.6e-4	2.0e-4	-9.39	-0.0917	✓
500	Direct $\rightarrow$ TR	+0.0169	[+0.009, +0.024]	0.0034	0.1955	0.0021	+2.64	+0.0127	
500	Direct $\rightarrow$ OMP+TR	-0.0720	[-0.092, -0.056]	0.0011	0.0631	0.0045	-4.43	-0.0610	
500	Direct $\rightarrow$ Shuf. TR	+0.0073	[-0.001, +0.014]	0.1386	1	0.1091	+1.04	+0.0031	
2,000	Direct $\rightarrow$ OMP	-0.0659	[-0.075, -0.057]	<1e-5	<1e-5	4.0e-5	-8.31	-0.0566	✓
2,000	Direct $\rightarrow$ Mean	-0.0505	[-0.058, -0.043]	<1e-5	<1e-5	3.9e-4	-7.53	-0.0392	✓
2,000	Direct $\rightarrow$ Random	-0.0980	[-0.105, -0.091]	<1e-5	<1e-5	2.0e-5	-14.86	-0.0816	✓
2,000	Direct $\rightarrow$ TR	+0.0030	[-0.003, +0.008]	0.3721	1	0.3755	+0.62	+0.0078	
2,000	Direct $\rightarrow$ OMP+TR	-0.0569	[-0.064, -0.050]	<1e-5	<1e-5	2.9e-4	-8.88	-0.0424	✓
2,000	Direct $\rightarrow$ Shuf. TR	-0.0008	[-0.006, +0.004]	0.8026	1	0.8254	-0.17	+0.0001	

selected conclusions; they do not convert the test split into a second discovery set.

G. The exact sign-flip floor

The paired t statistics behind the confirmation are large (up to $|t| = 61$, p down to 4×10^{-7}) because the five seed differences are unanimous and tightly clustered. Those p -values rest on the normal model. Under exact sign-flip inference, Eq. (17), five paired seeds admit 32 assignments and the smallest attainable two-sided p is $2/32 = 0.0625$. All 27 survivors attain that floor

on validation and on test, and all five seed differences point the same way in all 27 contrasts on both splits (Fig. 9): this is the maximum evidence a five-seed design can produce, and it cannot satisfy $p < 0.05$ without a distributional assumption. More seeds are the only direct remedy; additional post hoc tests on the same five runs are not.

H. Turkish source stage and degeneracy

Fourteen of the 30 XLM-15 Turkish source stages are degenerate—4 of 10 for TR, 6 of 10 for OMP+TR, 4 of 10

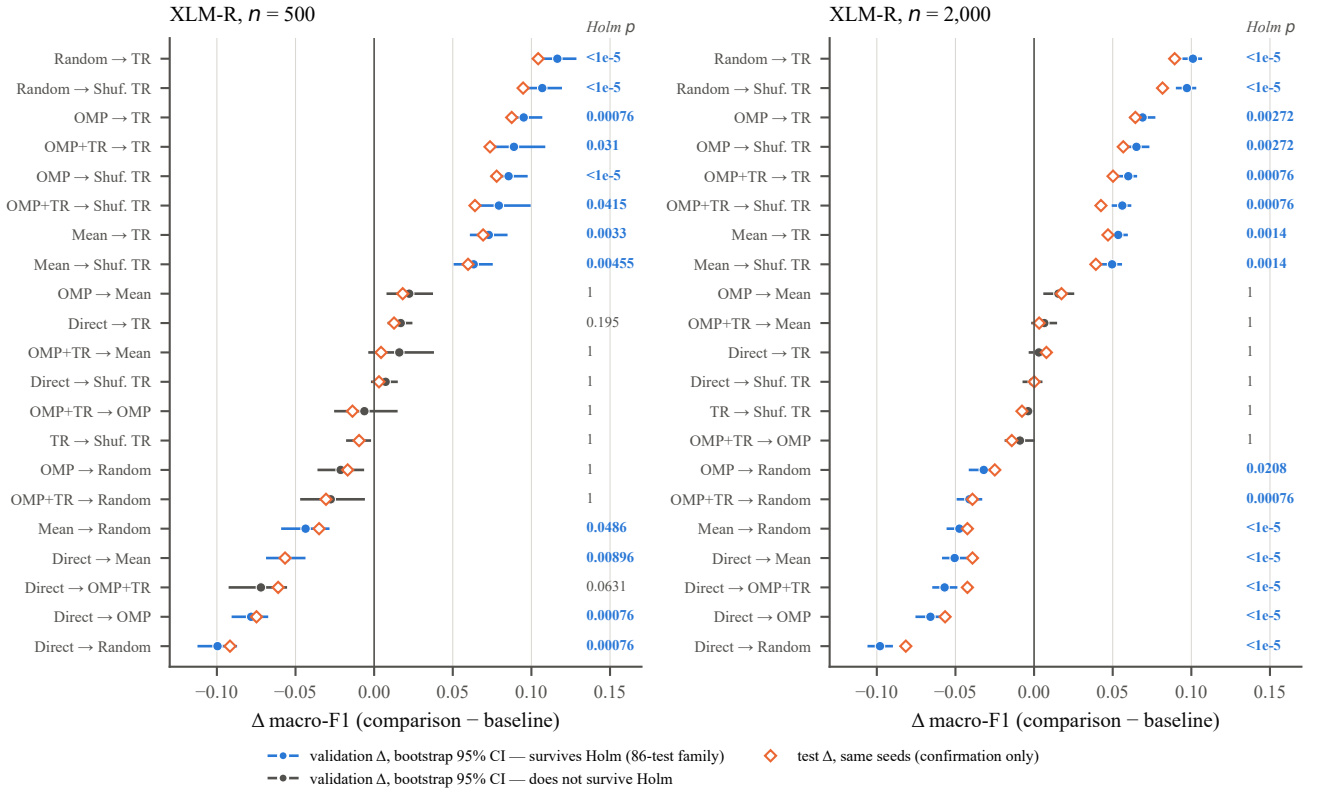


Fig. 8. All 42 within-base XLM-R contrasts (comparison minus baseline). Circles: conditional validation difference with 10,000-resample bootstrap 95% CI (results/stats.csv); blue if the contrast survives Holm correction over the 86-test family, grey otherwise, with the Holm-adjusted p printed at right. Hollow diamonds: paired test difference on the same five seeds (recomputed from results/runs/*.json; identical to T2/T5 of test_tables_received.md). Test values are confirmation only and were not used to select contrasts.

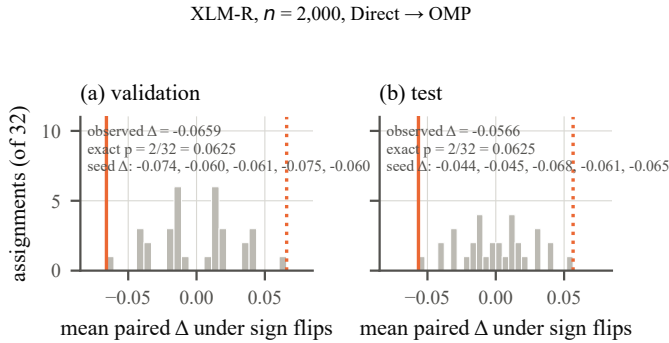


Fig. 9. Exact sign-flip distribution for the headline contrast (XLM-R, $n=2,000$, Direct $\rightarrow$ OMP), recomputed from the five per-seed differences in results/runs/*.json. The observed mean difference (solid line) and its mirror (dotted) are the two most extreme of the 32 assignments on both splits, so $p_{\text{exact}} = 2/32 = 0.0625$; the seed-level differences are printed in each panel.

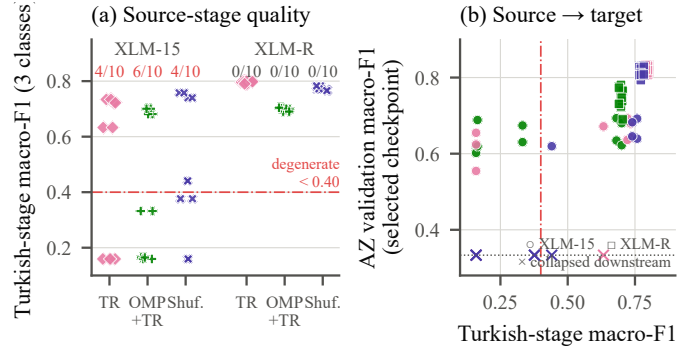


Fig. 10. Turkish source stage (results/aggregate.csv). (a) Turkish validation macro-F1 (three classes) of every Turkish-bearing run; the count is degenerate stages per cell. (b) Downstream Azerbaijani validation macro-F1 against source-stage macro-F1 for the 60 Turkish-bearing runs; crosses are runs that collapsed downstream.

for SHUF.TR (seven at each Azerbaijani size)—with three-class Turkish macro-F1 of 0.16–0.38, i.e., the source classifier predicted one or two classes; none of the 30 XLM-R stages is degenerate (Turkish macro-F1 0.69–0.79) (Fig. 10a). Downstream escape occurs after incompetent source stages as well as after competent ones (Fig. 10b), so source-stage and target-stage success are not interchangeable, and any statement that

Turkish training “stabilises” XLM-15 (5/5 escapes for OMP+TR at both sizes) must carry the degeneracy count. Table XVII collects the Turkish-related contrasts for both bases; on XLM-15 none approaches significance, and paired diagnostics are often unavailable because fewer than two seeds escaped in both arms.

TABLE XVII

TRUTH-RELATED CONTRASTS ON BOTH BASES (RESULTS/STATS.CSV). ESCAPE GIVES ESCAPED/DECLARED SEEDS IN BASELINE AND COMPARISON; Δ IS CONDITIONAL VALIDATION MACRO-F1 (COMPARISON MINUS BASELINE, SIGN-ADJUSTED WHEN THE STORED ORIENTATION IS REVERSED); "SHARED" IS THE NUMBER OF SEEDS ESCAPED IN BOTH ARMS; Δ TEST IS THE PAIRED TEST DIFFERENCE ON THOSE SEEDS, RECOMPUTED FROM RESULTS/RUNS/*.JSON.

Base	n	Contrast	Escape	Δ cond.	Welch p	Holm p	Shared	Paired p	Δ all-seed	Δ test	Surv.
XLM-R	500	Direct $\rightarrow$ TR	5/5, 5/5	+0.0169	0.0034	0.1955	5	0.0021	+0.0169	+0.0127	
XLM-R	500	Direct $\rightarrow$ Shuf. TR	5/5, 5/5	+0.0073	0.1386	1	5	0.1091	+0.0073	+0.0031	
XLM-R	500	Shuf. TR $\rightarrow$ TR	5/5, 5/5	+0.0096	0.0483	1	5	0.0091	+0.0096	+0.0096	
XLM-R	500	Direct $\rightarrow$ OMP+TR	5/5, 5/5	-0.0720	0.0011	0.0631	5	0.0045	-0.0720	-0.0610	
XLM-R	500	OMP $\rightarrow$ OMP+TR	5/5, 5/5	+0.0062	0.5952	1	5	0.5386	+0.0062	+0.0138	
XLM-R	500	OMP+TR $\rightarrow$ TR	5/5, 5/5	+0.0889	5.0e-4	0.0310	5	0.0013	+0.0889	+0.0737	✓
XLM-R	2,000	Direct $\rightarrow$ TR	5/5, 5/5	+0.0030	0.3721	1	5	0.3755	+0.0030	+0.0078	
XLM-R	2,000	Direct $\rightarrow$ Shuf. TR	5/5, 5/5	-0.0008	0.8026	1	5	0.8254	-0.0008	+0.0001	
XLM-R	2,000	Shuf. TR $\rightarrow$ TR	5/5, 5/5	+0.0038	0.0377	1	5	0.0635	+0.0038	+0.0077	
XLM-R	2,000	Direct $\rightarrow$ OMP+TR	5/5, 5/5	-0.0569	<1e-5	<1e-5	5	2.9e-4	-0.0569	-0.0424	✓
XLM-R	2,000	OMP $\rightarrow$ OMP+TR	5/5, 5/5	+0.0090	0.1127	1	5	0.1756	+0.0090	+0.0142	
XLM-R	2,000	OMP+TR $\rightarrow$ TR	5/5, 5/5	+0.0599	1.0e-5	7.6e-4	5	2.0e-5	+0.0599	+0.0502	✓
XLM-15	500	Direct $\rightarrow$ TR	3/5, 4/5	-0.0165	0.4797	1	3	0.7542	+0.0462	+0.0070	
XLM-15	500	Direct $\rightarrow$ Shuf. TR	3/5, 3/5	+0.0042	0.6784	1	2	0.5690	+0.0025	+0.0021	
XLM-15	500	Shuf. TR $\rightarrow$ TR	3/5, 4/5	-0.0207	0.3939	1	2	0.9701	+0.0437	-0.0038	
XLM-15	500	Direct $\rightarrow$ OMP+TR	3/5, 5/5	-0.0089	0.2924	1	3	0.4209	+0.1100	-0.0119	
XLM-15	500	OMP $\rightarrow$ OMP+TR	2/5, 5/5	+0.0945	0.4728	1	2	0.4734	+0.2108	+0.0915	
XLM-15	500	OMP+TR $\rightarrow$ TR	5/5, 4/5	-0.0076	0.7370	1	4	0.8427	-0.0638	-0.0039	
XLM-15	2,000	Direct $\rightarrow$ TR	2/5, 4/5	-0.0044	0.6597	1	2	0.3488	+0.1265	-0.0096	
XLM-15	2,000	Direct $\rightarrow$ Shuf. TR	2/5, 2/5	+0.0082	0.3605	1	0	–	-0.0049	–	
XLM-15	2,000	Shuf. TR $\rightarrow$ TR	2/5, 4/5	-0.0126	0.2440	1	1	–	+0.1315	-0.0185	
XLM-15	2,000	Direct $\rightarrow$ OMP+TR	2/5, 5/5	-0.0079	0.6001	1	2	0.8878	+0.1913	-0.0024	
XLM-15	2,000	OMP $\rightarrow$ OMP+TR	3/5, 5/5	+0.0564	0.5342	1	3	0.4193	+0.1689	+0.0701	
XLM-15	2,000	OMP+TR $\rightarrow$ TR	5/5, 4/5	+0.0035	0.8285	1	4	0.4582	-0.0648	-0.0079	

TABLE XVIII

MATCHED GENERALISATION GAP $F_{\text{test}} - F_{\text{val}}$ AT THE SELECTED CHECKPOINT (RESULTS/AGGREGATE.CSV).

Subset	N	Mean	Median	SD	Min	Max	$\Delta < 0$
All runs	164	-0.0126	-0.0131	0.0116	-0.0385	+0.0095	117
XLM-15	94	-0.0069	-0.0014	0.0101	-0.0348	+0.0095	48
XLM-R	70	-0.0202	-0.0202	0.0089	-0.0385	+0.0049	69
Escaped runs	130	-0.0159	-0.0163	0.0108	-0.0385	+0.0095	116
Collapsed runs	34	+0.0000	+0.0000	0.0018	-0.0076	+0.0069	1

I. Generalisation gap and validation–test agreement

Across all 164 restored checkpoints the matched gap $\Delta = F_{\text{test}}(t^*) - F_{\text{val}}(t^*)$ has mean -0.0126 , median -0.0131 , standard deviation 0.0116 , range -0.0385 to $+0.0095$, and is negative for 117 runs (Table XVIII, Fig. 11). XLM-R runs have a larger gap (mean -0.0202 , 69 of 70 negative) than XLM-15 runs (-0.0069), whose 34 collapsed runs contribute gaps near zero. A modest negative gap is the expected consequence of selecting the checkpoint on validation and supports treating the test split as confirmation rather than re-selection.

J. Fixed-step data-size sensitivity (XLM-15)

Table XIX extends the four non-Turkish XLM-15 conditions to $n = 10,000$ and $n = 20,914$ with three seeds. Escape is non-monotonic in n : DIRECT escapes 3/5, 2/5, 2/3 and 3/3; OMP 2/5, 3/5, 3/3 and 1/3; MEAN 2/5, 3/5, 1/3 and 3/3; RANDOM 3/5, 4/5, 1/3 and 1/3. At $n = 20,914$ the single escaped OMP seed reaches a conditional 0.7900 while the cell’s all-seed

value is 0.4856 , a direct example of selection-induced ranking distortion. Because every cell receives 2,000 steps, exposure falls from about 125 passes to 3 across the sweep; the table is a fixed-budget robustness probe, and cells with one escaped seed have only descriptive conditional values. The only measurable higher-data contrast, DIRECT $\rightarrow$ OMP at $n = 10,000$ ($+0.0230$, Welch $p = 0.112$), does not survive correction. H2 therefore receives no support in the form it was stated: the transplant does not help the encoder that lacks Azerbaijani pretraining, and on that encoder it is not distinguishable from its placebos.

K. Error taxonomy on test predictions

An automatic keyword taxonomy assigns each misclassified test example at $n = 2,000$, pooled over both bases and all five seeds (41,860 predictions per condition), to one of four categories (src/analysis/errors.py), in order of precedence: false friend (the text contains a word from a Turkish–Azerbaijani false-friend list,

ileresources/false_friends.csv), technical vocabulary (a short list of language-reform vocabulary pairs), morphological (the example’s tokens per word under the primary tokenizer exceed 4, a proxy for long suffix chains), or other (Table XX, Fig. 12). Error rates track the aggregate macro-F1: 28.0% for TR, 29.0% for OMP+TR, 31.5–31.6% for DIRECT, MEAN and SHUF. TR, 33.1% for RANDOM and 34.1% for OMP. The category shares are nearly identical across conditions (morphological 5.8–6.5%, technical 1.4–1.6%, false friend 0.2–0.3%, other 91.8–92.3%): the interventions change *how many* errors are made, not the keyword profile of the errors. Because the taxonomy

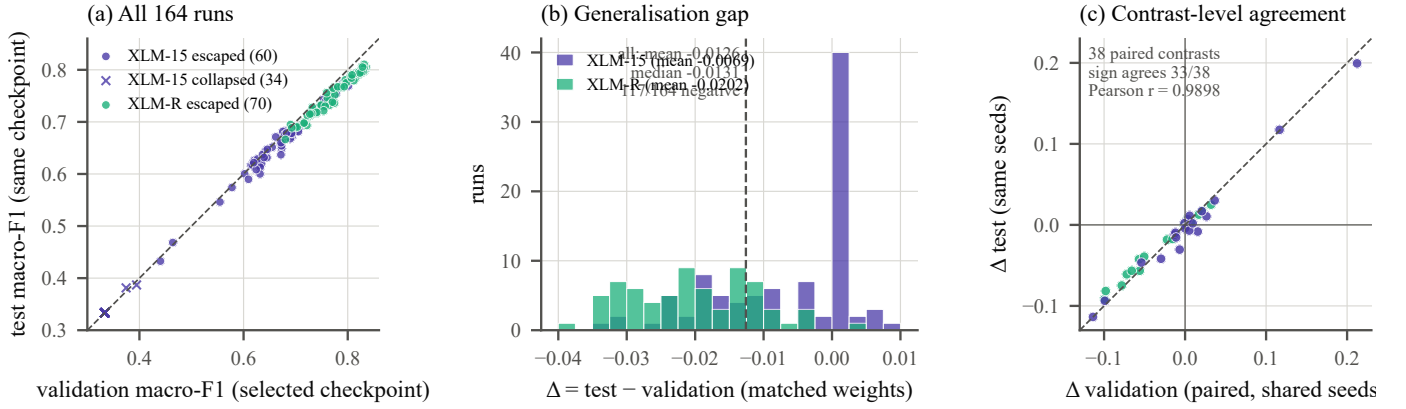


Fig. 11. Validation against test. (a) Selected-checkpoint validation macro-F1 versus test macro-F1 for all 164 runs (results/aggregate.csv). (b) Distribution of the matched gap. (c) Contrast-level agreement: paired validation difference versus paired test difference for the 38 contrasts with at least one shared escaped seed (T2 of test_tables_received.md).

TABLE XIX

XLM-15 FIXED-STEP SIZE SENSITIVITY (RESULTS/SUMMARY.CSV, T1 OF TEST_TABLES_RECEIVED.MD); FIVE SEEDS AT $n \leq 2,000$, THREE ABOVE.

n	Condition	Escape	Validation macro-F1		Test macro-F1	
			cond.	all-seed	cond.	all-seed
500	Direct	3/5	0.6306	0.5117	0.6229	0.5071
500	OMP	2/5	0.5272	0.4109	0.5248	0.4100
500	Mean	2/5	0.6359	0.4544	0.6302	0.4521
500	Random	3/5	0.6192	0.5049	0.6025	0.4949
2,000	Direct	2/5	0.6791	0.4799	0.6735	0.4790
2,000	OMP	3/5	0.6148	0.5023	0.6076	0.4980
2,000	Mean	3/5	0.6955	0.5507	0.6833	0.5434
2,000	Random	4/5	0.6699	0.6026	0.6529	0.5890
10,000	Direct	2/3	0.7427	0.6063	0.7253	0.5947
10,000	OMP	3/3	0.7657	0.7657	0.7438	0.7438
10,000	Mean	1/3	0.7653	0.4774	0.7539	0.4736
10,000	Random	1/3	0.6319	0.4533	0.6129	0.4444
20,914	Direct	3/3	0.7374	0.7374	0.7217	0.7217
20,914	OMP	1/3	0.7900	0.4856	0.7735	0.4801
20,914	Mean	3/3	0.7263	0.7263	0.7062	0.7062
20,914	Random	1/3	0.7671	0.4780	0.7417	0.4695

TABLE XX

AUTOMATIC ERROR TAXONOMY ON TEST PREDICTIONS AT $n = 2,000$, BOTH BASES AND ALL SEEDS POOLED (RESULTS/ERRORS.JSON). SHARES ARE PERCENTAGES OF ERRORS.

Condition	Evaluated	Errors	rate (%)	Share of errors (%)			
				Morph.	Techn.	False fr.	Other
Direct	41,860	13,176	31.48	6.5	1.4	0.2	91.8
OMP	41,860	14,286	34.13	6.0	1.6	0.3	92.1
Mean	41,860	13,221	31.58	5.9	1.5	0.3	92.3
Random	41,860	13,852	33.09	6.0	1.6	0.3	92.0
TR	41,860	11,736	28.04	6.2	1.5	0.3	92.0
OMP+TR	41,860	12,137	28.99	5.8	1.6	0.3	92.2
Shuf. TR	41,860	13,196	31.52	6.4	1.4	0.3	91.9

is lexical and 92% of errors fall outside its lists, we treat it as a null result about keyword-defined error types rather than as a characterisation of the errors; a 210-row manual sample (errors_manual_sample.csv) was exported but not annotated within the project window.

L. Compute cost per condition

Per-run wall-clock (Fig. 2, Table X) is governed by the encoder and the vocabulary: XLM-15 runs take 2.9–3.6 minutes and XLM-R runs 2.5–6.8 minutes, with the transplanted-vocabulary conditions slightly faster than the original-vocabulary ones because the tied output projection is smaller. The four data sizes differ little in wall-clock because the step budget is fixed. The whole grid consumed 8.36 process-hours on one GPU.

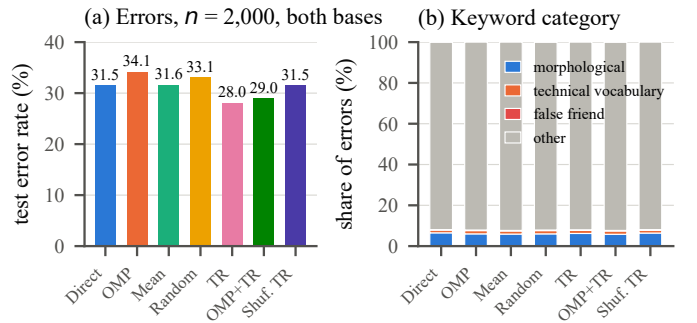


Fig. 12. Error taxonomy at $n = 2,000$ (results/errors.json). (a) Test error rate per condition, both bases and all seeds pooled. (b) Keyword category shares; “other” dominates in every condition.

VII. DISCUSSION

A. What the negative transplant result establishes

The cleanest finding is negative and implementation-bounded: the executed donor-tokenizer initialisation family damages XLM-R at $n = 500$ and $n = 2,000$. The conclusion triangulates across conditional and all-seed validation, restored-checkpoint test, unanimous seed direction, and three intrinsic probes (top-1 accuracy, BPC, norm calibration); bit-identical round trips and zero forward-logit deviation rule out serialisation drift. What the

design cannot say is *where* in the pipeline the damage originates. Functional anchors cover only 29.8% of the donor vocabulary, so 70% of the rows are reconstructions; the reconstruction is a 64-sparse combination of 256 cosine-nearest anchors solved in a space of a different dimension; the rows are then rescaled; the vocabulary, segmentation, truncation, tied output rows, and parameter count all change at once. Any of these could be the binding defect. Table XXI pairs each claim with its evidence and the qualifier it must carry.

Two observations narrow the space without resolving it. First, the three initialisers are ordered the same way on XLM-R by every measure— $\text{MEAN} > \text{OMP} > \text{RANDOM}$ on downstream macro-F1; $\text{OMP} > \text{MEAN} > \text{RANDOM}$ on BPC and top-1—so the fitted solve does carry some information beyond a random placement in the anchor span, but the crude mean of the nearest anchors is at least as good downstream. That pattern is more consistent with the anchor span itself being the limiting factor than with the solver. Second, the raw OMP reconstruction inflated XLM-R norms by $1.91\times$ but left XLM-15 norms nearly unchanged ($1.13\times$), which indicates that transferring coefficients solved on unit-normalised donor rows to unnormalised base rows interacts with the base model’s own norm distribution; rescaling fixes the scale but not the direction. Both observations are hypotheses for follow-up, not findings of this paper.

The result does not establish that OMP transplantation is generally harmful, that an Azerbaijani tokenizer cannot help, or that the precursor method fails on its own terms. It establishes that this implementation, on these encoders, with these anchors and this budget, does not recover the stronger original-tokenizer outcome on the encoder that already covers Azerbaijani.

B. H2 and the encoder without Azerbaijani exposure

H2 predicted that the transplant would help XLM-15 more than XLM-R. Descriptively, the cross-base difference of the $\text{DIRECT} \rightarrow \text{OMP}$ conditional effect at $n = 2,000$ is only 0.0017 (-0.0643 on XLM-15 vs. -0.0659 on XLM-R), and the escape-rate difference is $+0.2$ in favour of OMP on XLM-15 (3/5 vs. 2/5; Fisher $p = 1$). XLM-15’s tokenizer deficit is real and large (3.21 tokens per word; no diacritic tokens) and the donor removes it, yet nothing downstream improves within the measurement resolution, while the intrinsic probes show the transplanted masked-LM head is as blind to Azerbaijani as the original (2 of 1,502 vs. 2 of 2,323). The most economical reading is that segmentation efficiency is not the binding constraint for XLM-15 under this budget, or that whatever the better segmentation offers is cancelled by embedding rows the encoder cannot use. The design cannot separate these, and the frequent collapse of XLM-15 (60/94 escapes) means that most XLM-15 contrasts rest on two or three surviving seeds.

C. The Turkish stage: suggestive, unresolved

TR is numerically the best XLM-R condition at both sizes on validation and on test, its Direct-relative effect at $n = 500$ is 5.6 times that at $n = 2,000$, and the same-seed paired diagnostic at $n = 500$ is small ($p = 0.0021$). None of this survives

correction over the family it belongs to, and the design’s exact floor is 0.0625 regardless. The direction is compatible with Kardeş-NLU’s low-resource claim [6]; the magnitude, $+0.017$ macro-F1 at $n = 500$, is modest. The shuffled control adds a second suggestive but unresolved layer: real Turkish beats shuffled Turkish by $+0.010$ and $+0.004$, so if there is an order-sensitive component it is smaller than the total effect, and most of the Turkish benefit, if it exists, would be attributable to shared vocabulary and extra optimisation. SHUF.TR equals DIRECT to within 0.01, which means the shuffled source task on its own neither helps nor hurts measurably. On XLM-15, the Turkish stage cannot be assessed at all: 14 of 30 source stages produced a classifier that predicts one or two of the three Turkish classes, so the treatment was frequently not delivered, and excluding those runs would be an outcome-based exclusion.

D. Estimand choice under optimisation failure

XLM-15 illustrates why a single accuracy number misleads when fine-tuning collapses. Conditional means describe selected survivors and different conditions retain different seeds; an apparently high conditional score can coincide with a poor all-seed outcome (OMP at $n = 20,914$: 0.7900 vs. 0.4856). Escape probability, with five seeds, cannot distinguish 2/5 from 5/5 at conventional significance. Reporting all three estimands, never pooling them, and drawing the trajectories (Fig. 5) is the honest option, but a better design would predefine a joint hurdle endpoint—for example, all-seed macro-F1 with collapsed seeds charged at their score, which this paper reports but did not pre-register as primary—or would run enough seeds to estimate escape probability with useful precision. The fixed step budget, chosen to hold update opportunity constant across sizes, also means the study measures a fixed-budget regime rather than converged performance; the non-monotonic escape pattern across n (Section VI-J) is a property of that regime.

E. Relation to prior work

Our tokenizer diagnostics reproduce, for Azerbaijani, the fertility gap that Rust et al. identified as consequential for multilingual models [17], and the Turkish–Azerbaijani overlap lift is consistent with the lexical proximity that motivates Kardeş-NLU [6]. Unlike the embedding-initialisation literature [7]–[9], [11], which generally reports gains for the language whose tokenizer is installed, we observe a loss on the encoder that already covered the language and no gain on the encoder that did not, under a training-free initialisation and a short fine-tuning budget. The most plausible reconciliation is that those methods are usually followed by continued pretraining or larger adaptation budgets, which our protocol deliberately excluded to isolate the initialisation itself; our result is therefore about the initialisation, not about tokenizer adaptation with further training. The seed instability we document is in line with the fine-tuning-stability literature [28], [29], and our use of an explicit family, Holm correction, and an exact floor follows the recommendations of Dror et al. [31] and Bouthillier et al. [30].

TABLE XXI

CLAIM—EVIDENCE MAP. EVERY HEADLINE CLAIM, THE ARTEFACT THAT BACKS IT, AND THE QUALIFIER THAT MUST ACCOMPANY IT WHEREVER THE CLAIM IS REPEATED.

Claim	Evidence (table / figure; file)	Mandatory qualifier
The implemented OMP transplant lowers XLM-R macro-F1 at $n \in \{500, 2000\}$	Tables XIV, XV, XVI; Fig. 8; <code>stats.csv</code> , T5	Implementation-bounded; compound intervention; parametric survivor; exact floor 0.0625
Intrinsic probes corroborate the XLM-R failure	Table XIII; Fig. 4; <code>controls_*.json</code>	Diagnostics on 300 training lines; masked positions differ between tokenizers
Placebos order $\text{MEAN} > \text{OMP} > \text{RANDOM}$ downstream on XLM-R	Table XVI; Fig. 8	OMP→MEAN does not survive Holm; RANDOM contrasts do
Turkish stage is numerically best on XLM-R	Tables XIV, XV, XVII	Direct-relative gain does not survive Holm (0.1955 at $n = 500$); real-vs-shuffled unresolved
27 of 86 contrasts survive Holm, all on XLM-R; 27/27 confirmed on test	Table XXIII (Appendix); T5, T6	Model-based; exact sign-flip floor 0.0625 with five seeds
XLM-15 is dominated by collapse	Figs. 5, 6, 7; <code>aggregate.csv</code>	60/94 escapes; conditional and all-seed estimands diverge; no XLM-15 contrast survives
14 of 30 XLM-15 Turkish stages degenerate	Fig. 10; Table XVII	Reported, not excluded; treatment often not delivered
Transplant removes the XLM-15 segmentation deficit	Table XII; Fig. 3	Efficiency does not imply usable embeddings (H2 unsupported)
Test confirms validation (gap -0.0126 ; $r = 0.9898$)	Table XVIII; Fig. 11	Confirmation only; test never used for selection
Grid fits the compute envelope	Table X; Fig. 2	8.36 process-h on a rented H100; peak memory NOT MEASURED

VIII. LIMITATIONS AND THREATS TO VALIDITY

a) Construct validity: Tokenizer replacement jointly changes segmentation, vocabulary, embedding rows, tied output rows, truncation (1.257%→0.351% of examples for XLM-15), and parameter count. Word shuffling changes order, fluency, and source-task difficulty. Neither contrast isolates a single linguistic mechanism; Table II is the binding statement of what is identified. Five of the twelve (tokenizer, pre-stage) combinations—the placebos with a Turkish stage and the donor tokenizer with shuffled Turkish—were not run, so the interaction between transplant quality and the Turkish stage is observed only for OMP.

b) Model confounding: The bases differ in architecture, pretraining coverage, tokenizer scheme, and non-embedding capacity (1.8×). Cross-base differences cannot be attributed to Azerbaijani pretraining exposure alone, and the two encoders were not matched on any of these axes.

c) Optimisation validity: All cells receive the same 2,000-step budget despite dataset sizes spanning 42×, so the size sweep measures a fixed-budget regime. XLM-15 collapses in 34 of 94 runs, 14 of its 30 Turkish source stages are degenerate, and several conditional cells contain one or two observations. The experiment does not measure converged performance, and it does not measure tokenizer adaptation followed by continued pretraining.

d) Statistical validity: Only five (or three) seeds exist before escape conditioning. Exact paired inference cannot cross $p = 0.05$ with five seeds; every significance statement is model-based. Max-validation checkpoint selection and validation reporting use the same observations, which is why the test split is reported and why its gap is negative. Escape-rate contrasts have no power at this seed count. Holm control is applied to the pre-registered Welch family only; the paired, all-seed, and Fisher analyses are diagnostics.

e) Data and label validity: The upstream Azerbaijani revision is unpinning; the archived `data.tgz` is the only exact copy and the manifests are the only proof of identity. Pre-grid and grid splits differ by one example, so the human audit characterises the dataset, not the holdout. Unconditioned retained-label agreement is 65.67%, and the two annotators differ by nine points against gold; a non-trivial share of test labels is therefore uncertain, which bounds attainable macro-F1 and adds noise common to all conditions. Both datasets are user-generated sentiment text; the study does not speak to other domains or to formal Azerbaijani.

f) Test-protocol validity: The 164-entry ledger shows one test evaluation per run, at the verified restored checkpoint. Test-side inference is confirmation of validation-selected survivors, not discovery, and its paired tests inherit the five-seed exact floor.

g) Reproducibility and resources: The original lock did not describe the executed stack; two dependencies were undeclared; per-run weights and the Turkish cache are not distributed; peak GPU memory during the grid is NOT MEASURED; the grid ran on a rented H100 rather than the course workstation, in one stream. A clean-checkout reproduction on the course machine was not performed within the project window.

h) External validity: Evidence is limited to one binary sentiment dataset, one Turkish source task, one donor tokenizer, one sparse-reconstruction configuration ($k = 64$, $m = 256$, functional anchors), and two unmatched encoders. It does not establish transfer to other tasks, Azerbaijani varieties, donors, or architectures, and k and m were not ablated.

i) Licensing: LocalDoc is CC-BY-NC-SA-4.0, XLM-15 is CC-BY-NC-4.0, the HPLT donor is Apache-2.0, XLM-R is MIT, and the captured TRSAv1 card declares no licence. Redistribution rights must be checked per artefact; the non-commercial terms bind any downstream use of the dataset and of XLM-15-derived checkpoints.

IX. CONCLUSION AND FUTURE WORK

Under a pre-registered, fixed-budget protocol with 164 completed runs, the implemented OMP tokenizer-transplant pipeline consistently harms XLM-R at low and reference Azerbaijani data sizes—by 0.05–0.10 macro-F1 on validation and on the restored-checkpoint test split, in every seed, and with the ordering $\text{MEAN} > \text{OMP} > \text{RANDOM}$ among initialisers—while its intrinsic probes show that the reconstructed embedding rows do not preserve what the encoder’s masked-LM head needs. On XLM-15, which lacks Azerbaijani pretraining and has a severe segmentation deficit that the donor removes, the transplant produces no measurable improvement and is entangled with frequent optimisation collapse. Turkish intermediate training is directionally favourable for XLM-R, more so at $n = 500$, but is not statistically resolved against direct fine-tuning after correction over the primary family, and its order-sensitive component, if any, is smaller than its total effect. Twenty-seven contrasts survive Holm correction, all on XLM-R, and all twenty-seven keep their sign and significance on test; with five seeds, none can exceed the exact sign-flip floor of 0.0625.

The defensible contribution is a transparent comparison of adaptation pipelines with explicit failure-aware estimands, restored-checkpoint testing, multiplicity control, intrinsic diagnostics, source-stage accounting, and artefact-level disclosure of data provenance, software, compute, and test-set usage. The experiment does not show that Turkish transfer is ineffective, that an Azerbaijani tokenizer cannot help, that OMP transplantation is generally harmful, or that syntax has been isolated.

A stronger follow-up requires, in order of value: more independent seeds (the only remedy for the exact floor and for escape-rate power); a pre-registered joint hurdle endpoint combining failure probability and quality; an ablation of the anchor mode, k , m , and the norm-rescaling step, with an independently validated reference implementation of the transplant; a continued-pretraining arm after transplantation to separate initialisation from adaptation; gated or repeated Turkish source stages so that the treatment is always delivered; the five omitted (tokenizer, pre-stage) combinations; pinned data revisions and a clean-checkout reproduction on the course workstation with measured peak memory; and retention of selected checkpoints and predictions for error analysis beyond keyword taxonomies.

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a) *AI-assistance disclosure*: Large language models were used substantially in this project: for drafting and editing the paper text, for inspecting result artefacts and checking the consistency of numbers against files, for converting tables, for writing and debugging the analysis, figure, and table scripts in `report/ieee/`, and for parts of the experiment code

and its documentation. They did not generate experimental observations, alter result files, or train models. Every number in this paper was recomputed from the committed result files by the scripts named in the captions, and every claim, figure, and line of code is understood by and defensible by the authors. The experiments themselves ran with PyTorch [38], Transformers [39], scikit-learn [40], and the Hugging Face model and dataset hubs.

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APPENDIX A COMPLETE 36-CELL RESULTS

Table XXII lists every cell of the grid with escape rate and Wilson interval, conditional and all-seed validation macro-F1 with bootstrap intervals, test macro-F1 at the restored selected checkpoint, the number of seeds that escaped mid-training but ended collapsed, and the number of degenerate Turkish source stages.

APPENDIX B THE 27 HOLM-SURVIVING CONTRASTS AND THEIR TEST CONFIRMATION

Table XXIII lists the contrasts that survive Holm correction over the 86-test validation family, with the paired t test on test macro-F1 over the same five seeds and the Holm adjustment within the 27. All 27 are XLM-R contrasts; every seed difference has the same sign on both splits; the exact sign-flip p is 0.0625 for every row on both splits; mean shrinkage $|\Delta_{\text{test}}|/|\Delta_{\text{val}}|$ is 0.874.

APPENDIX C ALL PAIRED CONTRASTS

Table XXIV reports the paired validation and test differences on seeds that escaped in both arms for the 42 key contrasts of the companion table (38 with at least one shared seed), together with the all-seed differences, for both bases.

APPENDIX D FROZEN CONFIGURATION

The frozen configs/experiment.yaml (hash-locked in configs/CONFIG_HASHES.lock) fixes: seeds [42, 1337, 13, 7, 2024] and seeds_high_resource [42, 1337, 13]; deterministic: true; models FacebookAI/xlm-mlm-tlm-xnli15-1024 (primary), xlm-roberta-base (contrast), donor HPLT/hplt_bert_base_az at revision a126552c3073...; max_length 256; data LocalDoc/sentiments_dataset_azerbaijani (split train, hf_revision null, exclude_labels [neutral]) and maydogan/Turkish_SentimentAnalysis-TRSAv1 (max_examples 20000); split seed 12345 with val_ratio 0.10, test_ratio 0.15; train_sizes [500, 2000, 10000, 20914]; transplant method omp, k 64, n_candidates 256, normalize true, rescale_reconstructed true, seed 0; training lr 2.0e-5, max_steps 2000, epochs_tr 2,

TABLE XXII

ALL 36 CELLS (RESULTS/SUMMARY.CSV; TEST COLUMNS FROM T1 OF EVIDENCE/SUPPLEMENTAL/TEST_TABLES_RECEIVED.MD; BOTH EXACT AGGREGATES OF RESULTS/RUNS/*.JSON). WILSON AND BOOTSTRAP INTERVALS ARE 95%; "E→C" IS ESCAPED-THEN-TERMINAL-COLLAPSED SEEDS; "TR DEG." IS DEGENERATE TURKISH SOURCE STAGES (– FOR CONDITIONS WITHOUT A TURKISH STAGE).

Base	n	Condition	Escape		Val. cond.		Val. all-seed		Test cond.		Test		TR
			rate	Wilson	macro-F1	CI	macro-F1	CI	macro-F1	CI	all-seed	E→C	deg.
XLM-15	500	Direct	3/5	[0.23, 0.88]	0.6306	[0.621, 0.639]	0.5117	[0.391, 0.632]	0.6229	[0.618, 0.632]	0.5071	0	0
XLM-15	500	OMP	2/5	[0.12, 0.77]	0.5272	[0.440, 0.614]	0.4109	[0.333, 0.523]	0.5248	[0.433, 0.617]	0.4100	1	0
XLM-15	500	Mean	2/5	[0.12, 0.77]	0.6359	[0.627, 0.645]	0.4544	[0.333, 0.579]	0.6302	[0.629, 0.631]	0.4521	0	0
XLM-15	500	Random	3/5	[0.23, 0.88]	0.6192	[0.609, 0.632]	0.5049	[0.389, 0.621]	0.6025	[0.590, 0.618]	0.4949	0	0
XLM-15	500	TR	4/5	[0.38, 0.96]	0.6141	[0.575, 0.639]	0.5580	[0.439, 0.635]	0.6090	[0.569, 0.641]	0.5539	0	2
XLM-15	500	OMP+TR	5/5	[0.57, 1.00]	0.6217	[0.611, 0.630]	0.6217	[0.611, 0.630]	0.6142	[0.607, 0.619]	0.6142	0	3
XLM-15	500	Shuf. TR	3/5	[0.23, 0.88]	0.6348	[0.620, 0.646]	0.5143	[0.391, 0.638]	0.6336	[0.622, 0.647]	0.5135	0	2
XLM-15	2,000	Direct	2/5	[0.12, 0.77]	0.6791	[0.674, 0.684]	0.4799	[0.342, 0.618]	0.6735	[0.671, 0.676]	0.4790	0	0
XLM-15	2,000	OMP	3/5	[0.23, 0.88]	0.6148	[0.464, 0.691]	0.5023	[0.360, 0.645]	0.6076	[0.468, 0.687]	0.4980	0	0
XLM-15	2,000	Mean	3/5	[0.23, 0.88]	0.6955	[0.675, 0.706]	0.5507	[0.402, 0.700]	0.6833	[0.681, 0.687]	0.5434	0	0
XLM-15	2,000	Random	4/5	[0.38, 0.96]	0.6699	[0.664, 0.673]	0.6026	[0.467, 0.673]	0.6529	[0.641, 0.666]	0.5890	0	0
XLM-15	2,000	TR	4/5	[0.38, 0.96]	0.6747	[0.661, 0.687]	0.6064	[0.469, 0.683]	0.6663	[0.656, 0.677]	0.5997	0	2
XLM-15	2,000	OMP+TR	5/5	[0.57, 1.00]	0.6712	[0.645, 0.689]	0.6712	[0.645, 0.689]	0.6645	[0.644, 0.679]	0.6645	0	3
XLM-15	2,000	Shuf. TR	2/5	[0.12, 0.77]	0.6873	[0.682, 0.692]	0.4750	[0.333, 0.618]	0.6784	[0.678, 0.679]	0.4714	0	2
XLM-15	10,000	Direct	2/3	[0.21, 0.94]	0.7427	[0.740, 0.746]	0.6063	[0.333, 0.746]	0.7253	[0.724, 0.727]	0.5947	0	0
XLM-15	10,000	OMP	3/3	[0.44, 1.00]	0.7657	[0.749, 0.778]	0.7657	[0.749, 0.778]	0.7438	[0.730, 0.754]	0.7438	0	0
XLM-15	10,000	Mean	1/3	[0.06, 0.79]	0.7653	[0.765, 0.765]	0.4774	[0.333, 0.765]	0.7539	[0.754, 0.754]	0.4736	0	0
XLM-15	10,000	Random	1/3	[0.06, 0.79]	0.6319	[0.632, 0.632]	0.4533	[0.333, 0.632]	0.6129	[0.613, 0.613]	0.4444	0	0
XLM-15	20,914	Direct	3/3	[0.44, 1.00]	0.7374	[0.701, 0.758]	0.7374	[0.701, 0.758]	0.7217	[0.682, 0.743]	0.7217	0	0
XLM-15	20,914	OMP	1/3	[0.06, 0.79]	0.7900	[0.790, 0.790]	0.4856	[0.333, 0.790]	0.7735	[0.773, 0.773]	0.4801	0	0
XLM-15	20,914	Mean	3/3	[0.44, 1.00]	0.7263	[0.578, 0.801]	0.7263	[0.578, 0.801]	0.7062	[0.574, 0.775]	0.7062	1	0
XLM-15	20,914	Random	1/3	[0.06, 0.79]	0.7671	[0.767, 0.767]	0.4780	[0.333, 0.767]	0.7417	[0.742, 0.742]	0.4695	0	0
XLM-R	500	Direct	5/5	[0.57, 1.00]	0.7980	[0.794, 0.804]	0.7980	[0.794, 0.804]	0.7783	[0.773, 0.785]	0.7783	0	0
XLM-R	500	OMP	5/5	[0.57, 1.00]	0.7198	[0.710, 0.728]	0.7198	[0.710, 0.728]	0.7035	[0.693, 0.715]	0.7035	0	0
XLM-R	500	Mean	5/5	[0.57, 1.00]	0.7421	[0.732, 0.752]	0.7421	[0.732, 0.752]	0.7217	[0.716, 0.727]	0.7217	0	0
XLM-R	500	Random	5/5	[0.57, 1.00]	0.6985	[0.688, 0.709]	0.6985	[0.688, 0.709]	0.6867	[0.676, 0.694]	0.6867	0	0
XLM-R	500	TR	5/5	[0.57, 1.00]	0.8149	[0.810, 0.819]	0.8149	[0.810, 0.819]	0.7910	[0.788, 0.794]	0.7910	0	0
XLM-R	500	OMP+TR	5/5	[0.57, 1.00]	0.7261	[0.708, 0.741]	0.7261	[0.708, 0.741]	0.7173	[0.706, 0.727]	0.7173	0	0
XLM-R	500	Shuf. TR	5/5	[0.57, 1.00]	0.8054	[0.799, 0.810]	0.8054	[0.799, 0.810]	0.7814	[0.774, 0.787]	0.7814	0	0
XLM-R	2,000	Direct	5/5	[0.57, 1.00]	0.8277	[0.823, 0.833]	0.8277	[0.823, 0.833]	0.7970	[0.793, 0.801]	0.7970	0	0
XLM-R	2,000	OMP	5/5	[0.57, 1.00]	0.7618	[0.755, 0.769]	0.7618	[0.755, 0.769]	0.7404	[0.733, 0.749]	0.7404	0	0
XLM-R	2,000	Mean	5/5	[0.57, 1.00]	0.7772	[0.772, 0.783]	0.7772	[0.772, 0.783]	0.7578	[0.747, 0.767]	0.7578	0	0
XLM-R	2,000	Random	5/5	[0.57, 1.00]	0.7297	[0.725, 0.735]	0.7297	[0.725, 0.735]	0.7154	[0.714, 0.717]	0.7154	0	0
XLM-R	2,000	TR	5/5	[0.57, 1.00]	0.8307	[0.829, 0.832]	0.8307	[0.829, 0.832]	0.8048	[0.803, 0.808]	0.8048	0	0
XLM-R	2,000	OMP+TR	5/5	[0.57, 1.00]	0.7708	[0.766, 0.776]	0.7708	[0.766, 0.776]	0.7546	[0.748, 0.762]	0.7546	0	0
XLM-R	2,000	Shuf. TR	5/5	[0.57, 1.00]	0.8269	[0.825, 0.829]	0.8269	[0.825, 0.829]	0.7971	[0.795, 0.801]	0.7971	0	0

batch_size 32, grad_accum 1, warmup_ratio 0.1, weight_decay 0.01, fp16 true, early_stopping_patience null, eval_every_steps 200, eval_batch_size 64, metric macro_f1, metric_for_best_model eval_macro_f1, load_best_model_at_end true; the seven conditions of Table I; analysis_bootstrap_iters 10000, alpha 0.05, reference_size 2000; and run control priority_first true, skip_existing true, cache_turkish_stages true, contrast_sizes [500, 2000], parallel_streams 1, vram_ceiling_gb 10.0, vram_per_stream_gb 8.6. The escape threshold 0.40 is a code constant recorded in every run file (escape_threshold).

APPENDIX E

EVIDENCE MAP AND REPORTING CHECKLIST

Table XXV maps each class of claim to the file that backs it; all paths are relative to the repository root. The following qualifications are mandatory wherever these results are summarised: the grid completed 164/164 runs; conditional macro-F1 excludes failed seeds and is never pooled with all-seed macro-F1; all 27 primary Holm survivors are XLM-R contrasts; five-seed exact paired inference has a two-sided floor of 0.0625; 14/30 XLM-15 Turkish stages are degenerate; retained-label agreement is 65.67% without conditioning; the

pre-grid and grid splits differ by one example; and peak GPU memory is NOT MEASURED. Omitting any of these would materially overstate the evidence.

APPENDIX F

QUALITATIVE SEGMENTATION PROBES

Table XXVI shows how the three tokenizers segment seven Azerbaijani probe words (results/fertility.json): XLM-15 splits inflected words into 3–10 pieces, several of them single characters, and maps ə to an isolated piece; XLM-R and the donor keep most words in one or two pieces. These are illustrations of the fertility gap in Table XII, not corpus-level estimates.

TABLE XXIII
 VALIDATION SURVIVORS AND TEST CONFIRMATION (T5 OF
 EVIDENCE/SUPPLEMENTAL/TEST_TABLES_RECEIVED.MD;
 RESULTS/STATS.CSV).

<i>n</i>	Contrast	Δ val	Δ test	<i>t</i> (test)	<i>p</i> (test)	Holm/27
500	Direct $\rightarrow$ OMP	-0.0782	-0.0748	-10.71	4.30e-04	5.20e-03
500	Direct $\rightarrow$ Mean	-0.0560	-0.0566	-8.84	9.04e-04	8.54e-03
500	Direct $\rightarrow$ Random	-0.0996	-0.0917	-15.65	9.73e-05	2.04e-03
500	OMP+TR $\rightarrow$ TR	+0.0889	+0.0737	+10.47	4.70e-04	5.20e-03
500	OMP+TR $\rightarrow$ Shuf. TR	+0.0793	+0.0641	+8.97	8.54e-04	8.54e-03
500	OMP $\rightarrow$ TR	+0.0951	+0.0875	+12.26	2.54e-04	4.83e-03
500	OMP $\rightarrow$ Shuf. TR	+0.0855	+0.0779	+11.42	3.36e-04	5.07e-03
500	Mean $\rightarrow$ Random	-0.0436	-0.0350	-4.81	8.59e-03	1.72e-02
500	Mean $\rightarrow$ TR	+0.0729	+0.0693	+17.20	6.70e-05	1.61e-03
500	Mean $\rightarrow$ Shuf. TR	+0.0633	+0.0597	+13.56	1.71e-04	3.43e-03
500	Random $\rightarrow$ TR	+0.1165	+0.1043	+16.16	8.58e-05	1.89e-03
500	Random $\rightarrow$ Shuf. TR	+0.1069	+0.0947	+12.01	2.76e-04	4.96e-03
2,000	Direct $\rightarrow$ OMP+TR	-0.0569	-0.0424	-7.31	1.86e-03	1.08e-02
2,000	Direct $\rightarrow$ OMP	-0.0659	-0.0566	-10.92	4.00e-04	5.20e-03
2,000	Direct $\rightarrow$ Mean	-0.0505	-0.0392	-7.20	1.97e-03	1.08e-02
2,000	Direct $\rightarrow$ Random	-0.0980	-0.0816	-36.97	3.20e-06	8.31e-05
2,000	OMP+TR $\rightarrow$ Random	-0.0411	-0.0392	-8.09	1.27e-03	9.53e-03
2,000	OMP+TR $\rightarrow$ TR	+0.0599	+0.0502	+11.15	3.68e-04	5.15e-03
2,000	OMP+TR $\rightarrow$ Shuf. TR	+0.0561	+0.0425	+11.95	2.81e-04	4.96e-03
2,000	OMP $\rightarrow$ Random	-0.0320	-0.0250	-4.65	9.64e-03	1.72e-02
2,000	OMP $\rightarrow$ TR	+0.0690	+0.0644	+11.59	3.17e-04	5.07e-03
2,000	OMP $\rightarrow$ Shuf. TR	+0.0651	+0.0567	+17.02	6.99e-05	1.61e-03
2,000	Mean $\rightarrow$ Random	-0.0475	-0.0424	-6.93	2.28e-03	1.08e-02
2,000	Mean $\rightarrow$ TR	+0.0535	+0.0470	+8.22	1.19e-03	9.53e-03
2,000	Mean $\rightarrow$ Shuf. TR	+0.0496	+0.0393	+7.38	1.80e-03	1.08e-02
2,000	Random $\rightarrow$ TR	+0.1010	+0.0894	+61.23	4.26e-07	1.15e-05
2,000	Random $\rightarrow$ Shuf. TR	+0.0972	+0.0817	+34.30	4.31e-06	1.08e-04

TABLE XXIV

PAIRED AND ALL-SEED CONTRASTS, BOTH BASES (T2 OF EVIDENCE / SUPPLEMENTAL / TEST_TABLES_RECEIVED.MD). "N/A": NO SEED ESCAPED IN BOTH ARMS.

Base	<i>n</i>	Contrast	Δ val (paired)	Δ test (paired)	Shared	Same sign	Δ val (all-seed)	Δ test (all-seed)
XLM-R	500	Direct $\rightarrow$ TR	+0.0169	+0.0127	5	yes	+0.0169	+0.0127
XLM-R	500	Direct $\rightarrow$ OMP+TR	-0.0720	-0.0610	5	yes	-0.0720	-0.0610
XLM-R	500	Direct $\rightarrow$ Shuf. TR	+0.0073	+0.0031	5	yes	+0.0073	+0.0031
XLM-R	500	Direct $\rightarrow$ OMP	-0.0782	-0.0748	5	yes	-0.0782	-0.0748
XLM-R	500	Direct $\rightarrow$ Mean	-0.0560	-0.0566	5	yes	-0.0560	-0.0566
XLM-R	500	Direct $\rightarrow$ Random	-0.0996	-0.0917	5	yes	-0.0996	-0.0917
XLM-R	500	Random $\rightarrow$ OMP	+0.0213	+0.0168	5	yes	+0.0213	+0.0168
XLM-R	500	Mean $\rightarrow$ OMP	-0.0223	-0.0182	5	yes	-0.0223	-0.0182
XLM-R	2,000	Direct $\rightarrow$ TR	+0.0030	+0.0078	5	yes	+0.0030	+0.0078
XLM-R	2,000	Direct $\rightarrow$ OMP+TR	-0.0569	-0.0424	5	yes	-0.0569	-0.0424
XLM-R	2,000	Direct $\rightarrow$ Shuf. TR	-0.0008	+0.0001	5	NO	-0.0008	+0.0001
XLM-R	2,000	Direct $\rightarrow$ OMP	-0.0659	-0.0566	5	yes	-0.0659	-0.0566
XLM-R	2,000	Direct $\rightarrow$ Mean	-0.0505	-0.0392	5	yes	-0.0505	-0.0392
XLM-R	2,000	Direct $\rightarrow$ Random	-0.0980	-0.0816	5	yes	-0.0980	-0.0816
XLM-R	2,000	Random $\rightarrow$ OMP	+0.0320	+0.0250	5	yes	+0.0320	+0.0250
XLM-R	2,000	Mean $\rightarrow$ OMP	-0.0155	-0.0174	5	yes	-0.0155	-0.0174
XLM-15	500	Direct $\rightarrow$ TR	+0.0033	+0.0070	3	yes	+0.0462	+0.0468
XLM-15	500	Direct $\rightarrow$ OMP+TR	-0.0124	-0.0119	3	yes	+0.1100	+0.1071
XLM-15	500	Direct $\rightarrow$ Shuf. TR	-0.0008	+0.0021	2	NO	+0.0025	+0.0064
XLM-15	500	Direct $\rightarrow$ OMP	-0.0991	-0.0937	2	yes	-0.1008	-0.0971
XLM-15	500	Direct $\rightarrow$ Mean	+0.0056	+0.0113	1	yes	-0.0573	-0.0550
XLM-15	500	Direct $\rightarrow$ Random	-0.0297	-0.0418	1	yes	-0.0068	-0.0122
XLM-15	500	Random $\rightarrow$ OMP	n/a	n/a	0	–	-0.0940	-0.0849
XLM-15	500	Mean $\rightarrow$ OMP	-0.0128	-0.0120	1	yes	-0.0435	-0.0422
XLM-15	2,000	Direct $\rightarrow$ TR	-0.0119	-0.0096	2	yes	+0.1265	+0.1207
XLM-15	2,000	Direct $\rightarrow$ OMP+TR	+0.0022	-0.0024	2	NO	+0.1913	+0.1855
XLM-15	2,000	Direct $\rightarrow$ Shuf. TR	n/a	n/a	0	–	-0.0049	-0.0076
XLM-15	2,000	Direct $\rightarrow$ OMP	+0.0156	-0.0082	1	NO	+0.0224	+0.0189
XLM-15	2,000	Direct $\rightarrow$ Mean	+0.0264	+0.0106	2	yes	+0.0708	+0.0643
XLM-15	2,000	Direct $\rightarrow$ Random	-0.0067	-0.0305	2	yes	+0.1227	+0.1099
XLM-15	2,000	Random $\rightarrow$ OMP	-0.0539	-0.0465	3	yes	-0.1003	-0.0910
XLM-15	2,000	Mean $\rightarrow$ OMP	-0.0001	-0.0043	2	yes	-0.0484	-0.0454
XLM-15	10,000	Direct $\rightarrow$ OMP	+0.0206	+0.0170	2	yes	+0.1594	+0.1492
XLM-15	10,000	Direct $\rightarrow$ Mean	n/a	n/a	0	–	-0.1289	-0.1211
XLM-15	10,000	Direct $\rightarrow$ Random	-0.1137	-0.1136	1	yes	-0.1530	-0.1503
XLM-15	10,000	Random $\rightarrow$ OMP	+0.1166	+0.1174	1	yes	+0.3124	+0.2994
XLM-15	10,000	Mean $\rightarrow$ OMP	+0.0051	-0.0070	1	NO	+0.2883	+0.2702
XLM-15	20,914	Direct $\rightarrow$ OMP	+0.0365	+0.0302	1	yes	-0.2517	-0.2416
XLM-15	20,914	Direct $\rightarrow$ Mean	-0.0111	-0.0155	3	yes	-0.0111	-0.0155
XLM-15	20,914	Direct $\rightarrow$ Random	+0.0094	+0.0019	1	yes	-0.2594	-0.2522
XLM-15	20,914	Random $\rightarrow$ OMP	n/a	n/a	0	–	+0.0077	+0.0106
XLM-15	20,914	Mean $\rightarrow$ OMP	+0.2123	+0.1994	1	yes	-0.2406	-0.2261

TABLE XXV
EVIDENCE MAP.

Claim class	Backing files
Run completeness, timing	results/launcher_state.json; results/logs/*.log; run.log
Per-run observations	164 schema-v4 files in results/runs/; results/aggregate.csv
Cell summaries	results/summary.csv, summary.json, paper_tables.md
Inference family	results/stats.csv, stats.json (p_adjusted_holm, survives_family_ correction)
Test confirmation, exact floor	evidence/supplemental/ test_tables_received.md (T1-T6); results/test_evaluation_ ledger.jsonl
Splits, manifests, leakage	results/splits.json; data.tgz snapshot
Label audit (pre-grid)	evidence/carried_forward_ pre_grid/ (PROVENANCE.md, label_audit*.csv)
Anchors, transplants, controls	results/anchors.json; transplant__*.json; controls__*.json; embedding_norm_ report__*.json; cross_base_ transplant_quality.json; top1_accuracy.json
Tokenizer diagnostics	results/overlap_control.json; fertility.json; truncation.json; scramble.json; evidence/carried_forward_ pre_grid/diacritics.json
Error taxonomy	results/errors.json; errors_manual_sample.csv
Environment, compute	evidence/environment/ (ENVIRONMENT_AS_RUN.md, env_*_as_run.txt, regenerated pins)
Pre-registration	evidence/preregistration/ (experiment.yaml, FROZEN.md, CONFIG_ HASHES.lock)
Figures and tables of this paper	report/ieee/figures/make_*.py, report/ieee/tables/make_tables.py

TABLE XXVI

SEGMENTATION OF SEVEN AZERBAIJANI PROBE WORDS (RESULTS/FERTILITY . JSON); THE NUMBER OF PIECES IN PARENTHESES. □ MARKS A WORD-INITIAL SENTENCEPIECE MARKER; · MARKS THE FASTBPE END-OF-WORD MARKER.

Word	XLM-15 pieces (#)	XLM-R pieces (#)	AZ donor pieces (#)
<i>gəlmişdilər</i>	g ə l mis dil ə r· (7)	□gəlmiş dilər (2)	□gəlmişdi lər (2)
<i>qanunvericilik</i>	q anun ver ic ilik· (5)	□qanunvericilik (1)	□qanunvericilik (1)
<i>təyyarə</i>	t ə y yar ə· (5)	□təyyarə (1)	□təyyarə (1)
<i>kompyuter</i>	kom put er· (3)	□kompüter (1)	□kompüter (1)
<i>xəstəxanalarımızda</i>	x ə st ə x an alar ım ı z da· (10)	□xəstəxana larımızda (2)	□xəstəxana larımızda (2)
<i>əməkdaşlıq</i>	ə m ə k das lı q· (7)	□əməkdaşlıq (1)	□əməkdaşlıq (1)
<i>işləyirlər</i>	is l ə yir l ə r· (7)	□işləyir lər (2)	□işləyir lər (2)